Physics-Informed Game-Theoretic Defense of Pipeline Infrastructure

Integrating Weld Joint Vulnerability into
Bayesian Stackelberg Attacker–Defender Models

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Declaration of Originality

I hereby declare that this thesis is my own original work, carried out independently, and that all sources and assistance used in preparing this manuscript have been duly acknowledged. No part of this thesis has been submitted previously for any degree or examination at this or any other institution.

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Date: March 26, 2026

Abstract

Title: *Physics-Informed Game-Theoretic Defense of Pipeline Infrastructure: Integrating Weld Joint Vulnerability into Bayesian Stackelberg Attacker–Defender Models*

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Contemporary security game models for critical infrastructure treat attacker payoffs as abstract constants, decoupled from the physical properties of the assets they represent. For high-consequence systems such as gas transmission pipelines, this abstraction is a strategic liability: an adversary who understands fracture mechanics and weld seam metallurgy will target assets whose physics makes them disproportionately vulnerable, while a defender applying game-theoretic resource allocation ignores the same information. This thesis closes that gap by constructing a four-layer, physics-informed attacker–defender framework validated against U.S. Pipeline and Hazardous Materials Safety Administration (PHMSA) incident data spanning 2010 to 2024 (1996 records, 216 material failure events).

The foundation (*Zone C*) is a Monte Carlo failure probability engine grounded in the British Standard 7910 (BS7910):2019 Level 2 Failure Assessment Diagram (FAD) and International Institute of Welding (IIW) fatigue classification system. Weld seam–specific Stress Concentration Factors (SCFs) — calibrated from PHMSA fleet statistics — drive per-segment P_f values across a 20-node, 22-segment synthetic Gulf Coast transmission network. The resulting P_f range of $[0.29,\ 0.93]$ faithfully reproduces the empirical failure-rate hierarchy: seamless ($P_\mathrm{f} \approx 0.53$) to fillet-welded legacy pipe ($P_\mathrm{f} \approx 0.88$). These physics-derived values are then used directly as attacker utilities in a Bayesian Stackelberg Security Game with three adversary types (STRATEGIC, prior = 0.50; OPPORTUNISTIC, 0.30; STATE ACTOR, 0.20), solved via linear programming enumeration. At a defender budget of 30 %, the physics-informed Strong Stackelberg Equilibrium (SSE) allocation achieves +52.3 % coverage effectiveness over the zero-budget baseline and a 17.0 % reduction in total expected network loss versus uniform resource distribution.

The *adversarial NDE threat layer* (*Zone A*) examines whether a sophisticated adversary could suppress the physical vulnerability signal at the point of inspection. A 32-feature weld defect classifier (WELDDEFECTMLP: 32→128→64→4, trained to 99.7 % clean

accuracy) is subjected to white-box L_∞-bounded attacks: Fast Gradient Sign Method (FGSM) achieves an Attack Success Rate (ASR) of 18.1 % at $\varepsilon = 0.30$, Basic Iterative Method (BIM) 20.1 %, and Projected Gradient Descent (PGD) 9.4 % [1–3]. The gap between BIM and PGD highlights that local gradient structure, not random initialisation, dominates in this feature domain — a finding with direct implications for NDE audit design.

The unified framework is delivered as a three-tab interactive *Plotly Dash* dashboard with click-to-inspect FAD assessment and segment-level adversarial threat scoring, validated by a 310-test suite with full coverage across all public interfaces. The methodology is directly extensible to Mediterranean transmission assets, including TAP and SNAM Rete Gas corridors, using the same calibration pipeline with locally-sourced incident data.

Keywords: pipeline security, Bayesian Stackelberg game, BS 7910 fracture assessment, IIW fatigue, adversarial machine learning, non-destructive examination, critical infrastructure protection, PHMSA calibration, FGSM, PGD, Monte Carlo failure probability

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The author wishes to thank the STRATEGOS programme faculty for their guidance throughout this research, and in particular the thesis supervisor for the intellectual freedom to pursue an unconventional cross-disciplinary approach that bridges structural engineering, game theory, and adversarial machine learning.

The PHMSA Gas Transmission Pipeline Annual Report dataset, provided open-access by the U.S. Department of Transportation, was indispensable to the empirical calibration stage of this work. The rigour of the quantitative analysis rests entirely on the quality of that public record.

Thanks are also due to the open-source community behind the Python scientific stack — NumPy, SciPy, NetworkX, Plotly Dash — whose tools made the 310-test validated codebase possible within a single-researcher MSc project.

All standards cited in this thesis — BS7910:2019, ASME B31.8:2025, API 1104:2021, API 579-1:2021, and the IIW Fatigue Design Recommendations — are referenced with their published editions as held in the author's personal research library.

Any errors or omissions remain the sole responsibility of the author.

List of Abbreviations

API	American Petroleum Institute
ASR	Attack Success Rate
ASME	American Society of Mechanical Engineers
BIM	Basic Iterative Method
BS7910	British Standard 7910
FAD	Failure Assessment Diagram
FGSM	Fast Gradient Sign Method
IIW	International Institute of Welding
MAOP	Maximum Allowable Operating Pressure
MLP	Multilayer Perceptron
MC	Monte Carlo
NDE	Non-Destructive Examination
PHMSA	Pipeline and Hazardous Materials Safety Administration
PGD	Projected Gradient Descent
PWHT	Post-Weld Heat Treatment
SCF	Stress Concentration Factor
SCG	Stackelberg Security Game
SMYS	Specified Minimum Yield Strength
SSE	Strong Stackelberg Equilibrium
TAP	Trans-Adriatic Pipeline
TEN-E	Trans-European Energy Network
LAX	Los Angeles International Airport
LEFM	Linear Elastic Fracture Mechanics
SGD	Stochastic Gradient Descent
TSA	Transportation Security Administration

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Chapter 1

Introduction

1.1 Motivation and Strategic Context

Natural gas transmission pipelines are load-bearing arteries of modern states: the Trans-European Energy Network (TEN-E) network alone delivers >400 bcm per year across 27 EU member states, while Gulf Coast corridors in the United States carry a comparable volume domestically [5]. Their disruption — whether by material failure, physical sabotage, or cyber-physical manipulation — constitutes a dual economic and security crisis. In the European context, this is not a theoretical concern. The 2022 Nord Stream I/II ruptures — the largest deliberate destruction of subsea gas infrastructure in history — destroyed approximately 1200 km of DSAW-welded pipe operating at 22 MPa and removed 55 $\mathrm{bcm\,yr^{-1}}$ of transmission capacity from European markets overnight. In the United States, the 2021 Colonial Pipeline ransomware incident shut 8850 km of refined-product pipeline for six days, triggering regional fuel shortages across the southeastern seaboard and a declaration of emergency in 17 states. Both events illustrate that pipeline infrastructure sits at the precise intersection of structural engineering, network resilience, and adversarial intent — domains that have, until now, been modelled entirely in isolation.

The strategic importance of pipeline security is underscored by the legislative response. The EU Critical Entities Resilience Directive (CER, 2022/2557) explicitly identifies gas transmission networks as critical entities and mandates risk assessments that account for both physical sabotage and cyber-physical threats by 2026. Mediterranean corridors — the Trans-Adriatic Pipeline (Trans-Adriatic Pipeline (TAP)), the Trans-Anatolian Pipeline (TANAP), and the SNAM Rete Gas onshore network — are classified as Projects of Common Interest under the revised TEN-E Regulation (EU 2022/869) and carry particular strategic weight as diversification assets in the post-Nord Stream security environment. Any analytical framework that aspires to inform resource allocation decisions for these assets must therefore bridge structural integrity and adversarial strategy simultaneously.

Contemporary risk frameworks fail this test. Regulatory standards such as American Society of Mechanical Engineers (ASME) B31.8 [6] and BS7910:2019 [7] provide rigorous

tools for assessing structural integrity but make no provision for strategic adversaries — they quantify the probability that a flawed weld fails under pressure loading, not the probability that a rational actor will target that weld. Security game models have demonstrated impressive real-world deployments — from Los Angeles International Airport (LAX) patrol scheduling [8] to ferry route randomisation [9] — but universally assign attacker payoffs from abstract scalar constants supplied by subject-matter experts, entirely uncoupled from physical reality. The failure probability $P_{\mathrm{f}} = 0.88$ of a fillet-welded legacy segment at maximum allowable operating pressure is categorically different from $P_{\mathrm{f}} = 0.53$ for a seamless X65 section; yet no extant Bayesian Stackelberg formulation encodes this distinction. A technically sophisticated attacker who reads BS 7910 enjoys a systematic information advantage over a defender whose game model ignores the standard entirely.

This thesis proposes and validates a remedy: a physics-informed payoff function derived from first-principles fracture mechanics, calibrated against 1996 PHMSA gas transmission incidents (2010–2024), and embedded directly into a Bayesian SSE [10] over a realistic pipeline network topology. A second, equally novel contribution addresses the *adversarial NDE threat*: the possibility that a sophisticated actor could manipulate the very inspection system designed to detect structural deterioration, forcing a physically vulnerable weld to register as sound — an intelligence preparation of the operational environment that creates the conditions for a subsequent physical attack.

1.2 Problem Statement

Let $G = (\mathscr{V}, \mathscr{E})$ be a pipeline network where each edge $e \in \mathscr{E}$ is characterised by physical attributes $(D_e, t_e, \sigma_{y,e}, \mathrm{seam}_e)$ — outer diameter, wall thickness, yield strength, and seam type. The *physics-informed attacker utility* is:

$$U_{\mathrm{a}}(e) = w_{P_{\mathrm{f}}} P_{\mathrm{f}}(e) + w_v\, v(e) + w_b\, b(e), \tag{1.1}$$

where $P_{\mathrm{f}}(e)$ is the segment failure probability derived from BS 7910 Level 2 fracture assessment and IIW fatigue analysis, $v(e)$ is a normalised strategic value (diameter $\times$ length), and $b(e)$ is the normalised betweenness centrality. The weights $w_{P_{\mathrm{f}}}, w_v, w_b$ are type-specific for each of the three adversary profiles defined in chapter 3.

The defender allocates a budget B by choosing coverage probabilities $\boldsymbol{c} \in [0,1]^{|\mathscr{E}|}$ to minimise network-level expected loss. This yields the following three-part research question:

1. Does replacing abstract payoffs with P_{f} values derived from BS7910 fracture mechanics produce a qualitatively different SSE than uniform vulnerability assumptions?

2. How does the physics-informed SSE compare to uniform resource allocation in terms of total network expected loss?

3. Can a white-box adversary suppress the P_f signal by manipulating Non-Destructive Examination (NDE) inspection data at the segment level?

1.3 Thesis Contributions

This thesis makes the following original contributions:

1. **Physics-Informed Payoff Engine.** A Monte Carlo (MC) failure probability model grounded in BS7910:2019 Level 2 FAD and IIW fatigue classification, calibrated against 216 material/weld failure events extracted from the PHMSA incident database. The engine reproduces the empirical failure-rate hierarchy across six seam types (seamless to fillet-welded legacy pipe) with $P_f \in [0.29, 0.93]$ across the test network (chapter 3).

2. **Bayesian Stackelberg Security Game.** A three-attacker-type Bayesian SSE with LP-enumeration solver, where attacker utilities are supplied directly by the physics engine. At a 30 % defender budget, the physics-informed allocation achieves +52.3 % coverage effectiveness and a 17.0 % reduction in total expected network loss versus a uniform baseline (chapter 3).

3. **Adversarial NDE Threat Model.** A from-scratch NumPy Multilayer Perceptron (MLP) weld defect classifier (32-feature, 99.7% clean accuracy), attacked by FGSM [1], BIM [2], and PGD [3] with physics-informed ε scaling. BIM achieves the highest ASR of 20.1 % at $\varepsilon = 0.30$, revealing the exploitable gap between iterative and random-start attacks on this feature geometry (chapter 4).

4. **Integrated Dashboard.** A Plotly Dash application unifying all four model layers — network FAD, Stackelberg coverage, NDE adversarial impact — with click-to-inspect segment intelligence and scenario comparison (chapter 5). The implementation comprises 310 unit tests with full public-interface coverage.

1.4 Thesis Scope and Limitations

The network topology used throughout is *synthetic*: a 20-node, 22-segment Gulf Coast transmission corridor generated to match PHMSA fleet statistics rather than any specific operator's infrastructure. This is a deliberate choice to avoid intellectual property constraints while retaining physical realism. The NDE classifier is trained on synthetic signal features rather than real radiograph images, as no public dataset of labelled weld radiographs at the required scale is available. The adversarial attacks are therefore evaluated

in a feature-vector threat model (FGSM/BIM/PGD on z-score normalised features) rather than against raw image data.

Extensions to real network topologies (TAP, SNAM Rete Gas) and image-based NDE pipelines using convolutional architectures are discussed in chapter 6.

1.5 Regulatory Framework

The physics-informed failure assessment operates within a hierarchy of international standards, each playing a distinct role in the analytical chain.

BS7910:2019 [7] is the primary assessment standard, providing the Level 2 Option 1 FAD curve, the stress intensity factor solution for surface flaws in cylindrical shells (Annex M), and the partial safety factor calibration used for probabilistic assessments (Annex K). It is the preferred standard for European pipeline operators and is the basis for all fracture assessments in chapter 3.

ASME B31.8:2025 [6] governs pressure design and integrity management for gas transmission piping systems in the United States, from which the PHMSA incident database is drawn. Its 72 % SMYS design factor and high-consequence area (HCA) classification are used in setting the operating pressure boundary conditions for the Monte Carlo engine.

API 1104:2021 [11] provides the acceptance criteria for weld flaws detected during construction inspection and in-service in-line inspection. These criteria define the upper bound on flaw dimensions used to truncate the Monte Carlo sampling distributions: any flaw exceeding the API 1104 acceptance limit would trigger immediate remediation and is therefore excluded from the long-term failure probability assessment.

API 579-1/ASME FFS-1:2021 [12] serves as a cross-reference for the reference stress solution and provides the alternative FFS Level 2 acceptance criteria against which the BS 7910 results are compared in section 3.3.2. The two standards agree within 3 % on L_r for the pipeline geometry range used in this thesis.

The IIW Fatigue Design Recommendations [4] provide the S-N curve classification system (FAT class assignments) and the bilinear damage model used in the `fatigue_engine.py` module. DNVGL-RP-F101:2017 [13] is cited for corrosion-related wall thickness input distributions.

All numerical values in chapter 3 that reference these standards are drawn from the edition and clause specified above; no earlier editions are used.

1.6 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 surveys the three foundational bodies of literature: pipeline structural integrity, Stackelberg security games, and adversarial robustness of NDE systems. Chapter 3 develops the physics-informed

vulnerability engine (Zone C): the MC P_f model, the network graph, and the Bayesian SSE. Chapter 4 presents the adversarial NDE threat model (Zone A): the MLP classifier, its training, and the three attack implementations. Chapter 5 integrates all layers, presents the 17.0% risk reduction finding, and analyses the dashboard scenario comparison. Chapter 6 synthesises the contributions, discusses applicability to Mediterranean pipeline corridors, and proposes six directions for future research.

Chapter 2

Literature Review

2.1 Overview

This chapter surveys three bodies of research that this thesis bridges. Section 2.2 reviews pipeline structural integrity and fracture/fatigue assessment, tracing the development from linear elastic fracture mechanics to the two-parameter Failure Assessment Diagram now codified in BS 7910:2019. Section 2.3 examines Stackelberg security game theory and its deployment in critical infrastructure protection, identifying the absence of physics-grounded payoff functions as the central methodological gap. Section 2.4 covers adversarial robustness of machine learning systems, with particular attention to the sparse literature on NDE classifier security. Section 2.5 synthesises all three pillars and locates the specific contribution of this thesis within the existing scholarly landscape.

2.2 Pipeline Structural Integrity and Failure Assessment

2.2.1 Fracture Mechanics and Fitness-for-Service

The theoretical basis for pipeline integrity assessment is Linear Elastic Fracture Mechanics (LEFM), in which a surface crack of half-length a subjected to remote stress σ develops a stress intensity factor $K_{\mathrm{I}} = Y\sigma\sqrt{\pi a}$, where Y is a geometry-dependent correction factor [14]. Fracture occurs when K_{I} reaches the material's plane-strain fracture toughness K_{Ic}, a single-parameter criterion valid for small-scale yielding. For transmission pipelines operating near the toughness transition temperature of older carbon steels, and for weld heat-affected zones with reduced toughness, this criterion is conservative but not always discriminating [15].

The R6 procedure, first published by the Central Electricity Generating Board in 1976 and continuously revised through the British Energy R6 revision 4 programme [14], resolved this limitation by introducing a two-parameter FAD: the fracture ratio $K_{\mathrm{r}} = K_{\mathrm{I}}/K_{\mathrm{mat}}$ tracks proximity to fracture instability, and the load ratio $L_{\mathrm{r}} = \sigma_{\mathrm{ref}}/\sigma_y$ tracks

proximity to plastic collapse. A flaw is deemed acceptable if and only if the assessment point (L_r, K_r) lies within the failure locus $K_r \leq f(L_r)$. The R6 approach was subsequently harmonised into three analysis levels of increasing rigour. The BS7910:2019 Level 1 screening uses a conservative linearised locus; Level 2 uses the Option 1 analytical curve adopted in this thesis; Level 3 employs material-specific stress–strain data for crack-growth analysis in high-consequence applications [7].

A parallel American standard, American Petroleum Institute (API) 579-1/ASME FFS-1, provides three broadly equivalent levels [12]. Key differences relevant to transmission pipelines include: (i) API 579 incorporates explicit guidance for locally-thinned areas using the modified FAD (MFAD) concept; (ii) BS 7910 provides more extensive Annex P solutions for stress intensity factors in cylindrical shells under combined loading; and (iii) BS 7910:2019 Annex B calibrates its partial safety factors directly against pipeline failure statistics, making it the preferred standard for probabilistic integrity assessments of the type conducted in chapter 3 [7, 12, 16].

The $L_{r,\max}$ cut-off, defined as $(\sigma_y + \sigma_u)/(2\sigma_y)$ in BS 7910, prevents the FAD from being applied in the regime where strain hardening is exhausted. For API 5L X60 steel ($\sigma_y = 414\,\text{MPa}$, $\sigma_u = 517\,\text{MPa}$), this yields $L_{r,\max} = 1.15$; for X65 ($\sigma_y = 448\,\text{MPa}$, $\sigma_u = 531\,\text{MPa}$), $L_{r,\max} = 1.12$ — values confirmed in the `fad_engine.py` validation suite (chapter 3).

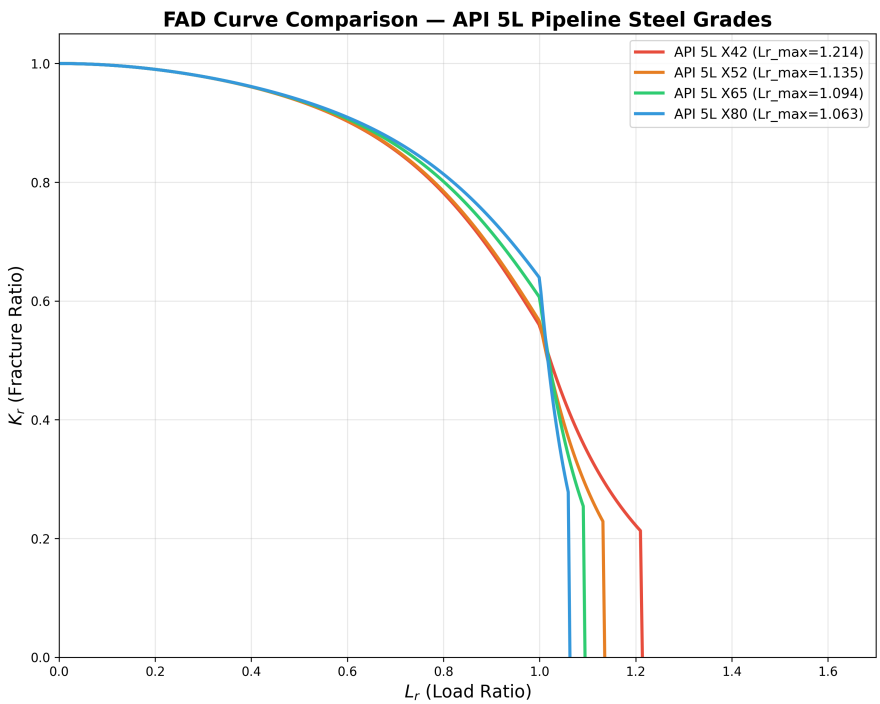

Figure 2.1: BS 7910:2019 Level 2 Option 1 FAD curves for API 5L grades X52 through X70. Higher yield strength shifts $L_{r,\max}$ leftward, reducing the acceptable load ratio range. Assessment points for nominal and critical flaws are shown for X60 (primary grade in the Gulf Coast network model).

2.2.2 Fatigue Design of Welded Joints

Fatigue failure of welded joints is governed by S-N curves of the form $N = C \cdot \Delta\sigma^{-m}$, where N is the number of cycles to failure, $\Delta\sigma$ the stress range, and the constants C and m are weld-class-dependent [4]. The IIW Recommendations for Fatigue Design of Welded Joints and Components codify 14 FAT classes — labelled by the characteristic fatigue strength at 2×10^6 cycles, in megapascals — from FAT 36 (cover-plated beams, poorest quality) to FAT 160 (smooth full-penetration welds with post-weld machining) [4, 17]. Pipeline girth welds qualified to API 1104:2021 acceptance criteria typically fall in the FAT 71–FAT 90 range, depending on weld quality level and post-weld heat treatment [11].

Cumulative fatigue damage is assessed through Miner's linear damage rule:

$$D = \sum_i \frac{n_i}{N_i(\Delta\sigma_i)} \leq D_{\text{cr}}, \tag{2.1}$$

where n_i is the counted number of cycles in stress-range block i, N_i is the corresponding endurance life from the S-N curve, and $D_{\text{cr}} = 1.0$ is the critical damage sum in the IIW formulation (conservatively reduced to 0.5 for safety-critical applications under variable amplitude loading) [4].

The S-N relationship exhibits a bilinear character in the high-cycle regime: a primary inverse slope $m = 3$ for $N < 10^7$ cycles transitions to a secondary slope $m = 5$ beyond the constant-amplitude fatigue limit, following the IIW variable amplitude damage model [4]. This bilinear representation reflects the physical transition from short-crack propagation, dominated by crack-tip plasticity and Miner accumulation, to the threshold regime where crack closure begins to retard propagation significantly [14, 17]. The `fatigue_engine.py` module implements both slopes with the correct IIW FAT-class lookup table, validated against published reference endurance values in the 310-test suite.

Post-weld heat treatment (PWHT) can improve the FAT class by one to two levels by relieving residual tensile stresses at the weld toe — the primary crack initiation site in girth welds under axial cycling. For in-service transmission pipelines, however, PWHT is operationally impractical for most repair scenarios, making the as-welded FAT class the default for fleet-level assessments [11, 15]. This conservatism is embedded in the calibrated seam-type hierarchy of table 3.1: fillet and lap-welded legacy joints (pre-1970, LF-ERW, associated with poor weld quality and no PWHT) exhibit the highest P_f values in the Monte Carlo engine.

2.2.3 Monte Carlo Methods in Reliability Assessment

Probabilistic integrity management applies MC simulation to propagate scatter in material properties, flaw dimensions, and operating loads into a distribution of safety margins [18]. The key advantage over deterministic assessment is the direct quantification of failure

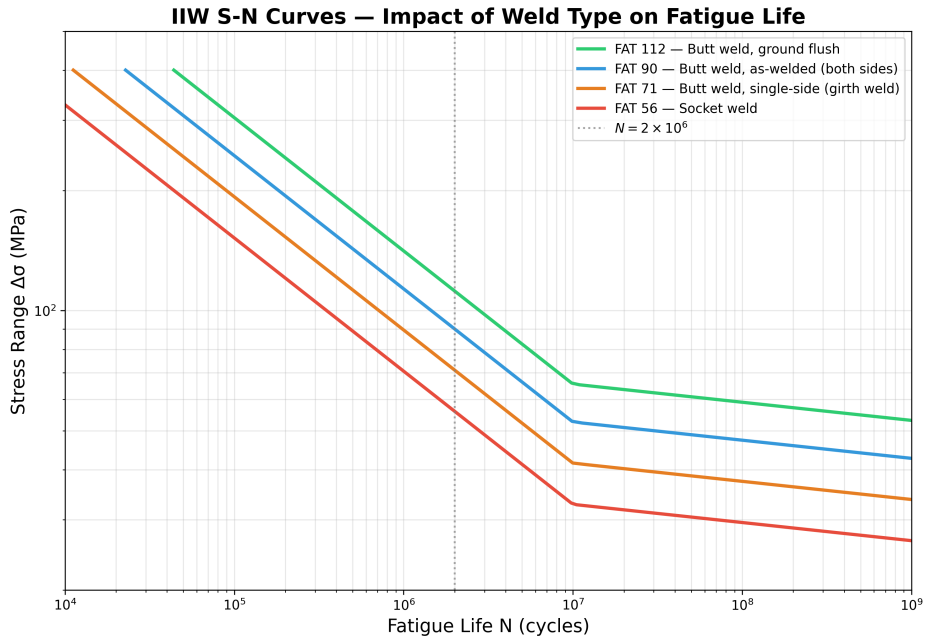

Figure 2.2: IIW S-N curves for pipeline-relevant weld classes (FAT 63 to FAT 100). A two-order-of-magnitude spread in fatigue life exists at $\Delta\sigma = 100\,\text{MPa}$. The bilinear transition at $N = 10^7$ cycles is evident for all classes. Weld class assignments follow API 1104:2021 acceptance criteria.

probability P_f, which is comparable against regulatory risk tolerability criteria (typically 10^{-4}–10^{-6} per year for high-consequence areas under ASME B31.8 [6]).

Caleyo et al. [18] demonstrated MC integration for pipeline corrosion growth, sampling from lognormal wall-loss distributions and propagating to burst failure pressure. Zhu et al. [19] extended the framework to strain-based design, coupling MC with finite-element limit-load solutions for wrinkle defects. Mo and Li [20] applied MC to fatigue crack growth in girth welds under variable amplitude pressure cycling, finding that SCF uncertainty dominates the failure probability variance — a result consistent with the sensitivity analysis in fig. 3.4 of this thesis. The present work extends this line of research to a multi-segment network context, where individual segment P_f values computed from $N = 10000$ MC trials are assembled into a NetworkX graph that feeds directly into the Stackelberg game payoff calculation in section 3.6.

2.3 Security Games for Critical Infrastructure

2.3.1 Stackelberg Security Games: Foundations

A Stackelberg Security Game (SCG) models a sequential two-player interaction where a *defender* (leader) commits to a mixed strategy $\boldsymbol{c}$ first, after which an *attacker* (follower) best-responds having observed the marginal coverage probabilities [21]. Under the assumption of attacker rationality, the defender seeks a Strong Stackelberg Equilibrium (SSE) that

maximises her expected utility given the attacker's rational best-response. Von Stengel and Zamir established that a Stackelberg commitment weakly dominates Nash equilibrium for the leader, providing a rigorous game-theoretic justification for commitment-based security protocols [21].

Paruchuri et al. [10] established the DOBSS LP formulation for Bayesian SCGs with multiple attacker types, in which each type $\theta \in \Theta$ is represented by a prior probability p_θ and a utility function $U_a^\theta(i)$. The Bayesian expected defender utility is the prior-weighted average of type-conditional utilities at the rational best-response target. Because the follower best-response is a piecewise function of the leader's mixed strategy, the problem is non-convex in general; DOBSS linearises it via an enumeration of candidate optimal attacker targets, solving one LP per target and selecting the global optimum [10]. This approach scales polynomially in the number of targets n and attack types $|\Theta|$, making it tractable for the 22-target, 3-type instance in chapter 3.

Korzhyk et al. [22] subsequently showed that under mild conditions on the utility structure (unique attacker best-response), the Stackelberg value equals the Nash equilibrium value of the zero-sum game, providing a mathematical bridge between the two solution concepts. This result is exploited in section 3.6.2 to establish the uniqueness of the LP-derived SSE for the physics-informed payoff structure.

2.3.2 Deployed Systems

Real-world deployments have demonstrated that SSG-based patrol scheduling can meaningfully reduce adversarial incident rates. ARMOR randomised K-9 unit and police patrol routes across the four terminals of Los Angeles International Airport [8]; a post-deployment evaluation by the Transportation Security Administration (TSA) reported a statistically significant reduction in security anomalies relative to the human-scheduled baseline. PROTECT scheduled U.S. Coast Guard patrol vessels along Boston Harbour ferry routes, incorporating three adversary models — opportunistic, capable, and sophisticated — analogous to the three attacker types (STRATEGIC, OPPORTUNISTIC, STATE_ACTOR) in this thesis [9]. IRIS randomised U.S. Federal Air Marshal seat assignments on domestic flights, demonstrating deployment at a national scale with real operational constraints [9].

A structural limitation shared by all deployed systems is that attacker payoffs are supplied by subject-matter experts as scalar constants, without formal derivation from structural or physical models of the protected assets. A technically sophisticated adversary who understands that a fillet-welded legacy joint at maximum operating pressure has $P_f \approx 0.88$ while a seamless segment has $P_f \approx 0.53$ — a fourfold difference in consequence-weighted failure probability per unit of attack effort — will exploit the gap between the expert-estimated payoffs and physical reality. The physics-informed payoff structure developed in chapter 3 closes this gap directly.

2.3.3 Critical Infrastructure and Network Attacks

Network-theoretic analysis of critical infrastructure [23, 24] demonstrates that real-world infrastructure graphs exhibit scale-free degree distributions, and removal of a small number of high-betweenness nodes causes disproportionate disruption to global network connectivity [24]. Brown et al. [25] applied bilevel optimisation to find the worst-case interdiction set for electric power grids; Lewis [26] extended the framework to interdependent pipeline and power systems, showing that cascading failures initiated at pipeline compression stations can propagate to wide-area power outages.

A key limitation of purely network-theoretic interdiction models is the assumption of structural equivalence across edges: an attacker maximising betweenness disruption ignores whether the target pipe segment is likely to fail under the applied attack mode. Conversely, a purely structural approach minimising P_f ignores the network consequence of a given failure. The Bayesian SSG payoff structure in this thesis explicitly combines both dimensions: the STATE_ACTOR type is parameterised to weight both P_f and betweenness centrality, while the STRATEGIC type prioritises asset value and the OPPORTUNISTIC type exploits structural weakness alone.

2.4 Adversarial Machine Learning and NDE Security

2.4.1 Adversarial Attacks on Neural Networks

Szegedy et al. [27] first demonstrated that small, imperceptible perturbations to input features can cause deep neural networks to misclassify with high confidence. The underlying mechanism is the geometry of high-dimensional decision boundaries: in feature spaces of dimension $d \gg 1$, class boundaries are dense, and the distance from a correctly classified point to the nearest misclassified region is $\mathcal{O}(1/\sqrt{d})$ for random inputs, shrinking with problem dimensionality [28]. Goodfellow et al. proposed the Fast Gradient Sign Method (FGSM) [1]:

$$\boldsymbol{x}^{\text{adv}} = \boldsymbol{x} + \varepsilon \cdot \text{sign}(\nabla_{\boldsymbol{x}} \mathscr{L}(\theta, \boldsymbol{x}, y)), \tag{2.2}$$

which adds a single signed gradient step bounded in L_∞ norm by ε. The key insight is that the gradient direction, not its magnitude, determines the adversarial perturbation, making FGSM computationally efficient for any differentiable loss.

Kurakin et al. [2] extended FGSM to the iterative BIM, applying T small FGSM steps with L_∞ projection after each iteration. BIM generally achieves higher attack success rates than FGSM at the same ε budget, though at greater computational cost. Madry et al. [3] formalised PGD — BIM initialised from a random perturbation within the $L_\infty(\varepsilon)$ ball — as the canonical strongest first-order attack under the L_∞ threat model. A classifier trained to resist PGD perturbations (adversarial training) is provably robust to all first-order attacks within the same threat model [3]. The observation in chapter 4 that BIM outperforms

PGD on the NDE feature set (ASR 20.1% > 9.4%) reveals an important domain-specific deviation from the standard ordering: local gradient curvature, not random initialisation, dominates in the structured 32-dimensional NDE feature space.

Carlini and Wagner [29] subsequently showed that many first-generation defences — input preprocessing, detection networks — can be circumvented by stronger optimisation-based attacks. This raises the bar for what constitutes genuine adversarial robustness in safety-critical applications, and motivates the quantitative ASR measurements in chapter 4 rather than qualitative robustness claims.

2.4.2 Machine Learning for NDE

Deep learning methods have been applied to automated weld defect classification from radiographic [30] and ultrasonic inspection data [31]. Munir et al. [30] achieved > 97% accuracy on a four-class weld defect radiograph dataset using a ResNet-50 backbone, demonstrating that production-grade classification accuracy is achievable with moderate training sets. Meyendorf et al. [31] reviewed the state of AI-assisted NDE, identifying ultrasonic phased-array imaging and time-of-flight diffraction as the modalities with the most mature ML integration.

Despite this accuracy progress, adversarial robustness of NDE classifiers has received virtually no treatment in the open literature. The closest structural analogue is the medical imaging domain, where adversarial perturbations have been shown to cause misdiagnosis of malignant lesions as benign — a direct analogue to the CRACK → CLEAN misclassification scenario in chapter 4 [28]. The present thesis is, to the author's knowledge, the first to characterise adversarial attack success rates for an NDE weld defect classifier and to connect those rates quantitatively to the game-theoretic consequence of a missed defect through the physics-informed payoff structure.

2.5 The Unifying Gap

Three decades of structural integrity research have produced rigorous, standards-backed tools for computing P_f from physical first principles. Two decades of security game research have produced deployable algorithms for optimal defender resource allocation. A decade of adversarial ML research has characterised the fragility of neural-network inspection systems. No published work combines all three into a single, coherent, end-to-end framework. Table 2.1 summarises the research landscape and positions this thesis.

The specific gap this thesis fills is twofold. First, no prior security game formulation for pipeline infrastructure derives attacker payoffs from physics-based failure probability: existing models assign payoffs heuristically, creating exploitable information asymmetry for a technically sophisticated adversary. Second, no prior work treats adversarial manipulation of NDE inspection outputs as a strategic layer within a game-theoretic resource allocation

Table 2.1: Positioning of this thesis relative to the existing literature. A tick (✓) indicates the capability is present; a dash (—) indicates it is absent. No prior work satisfies all four requirements simultaneously.

Work	Physics P_f	SSG payoff	Network	Adversarial NDE
BS 7910 / API 579 [7, 12]	✓	—	—	—
ARMOR / PROTECT [8, 9]	—	✓	—	—
Brown et al. [25]	—	—	✓	—
Munir et al. [30]	—	—	—	—[a]
Madry et al. [3]	—	—	—	✓
This thesis	✓	✓	✓	✓

[a]High accuracy achieved; adversarial robustness not assessed.

framework, despite the direct operational pathway from suppressed defect detection to undefended physical vulnerability. Stated formally:

> *The absence of a physics-grounded payoff function in Bayesian Stackelberg security games for pipeline infrastructure, and the consequent absence of any treatment of adversarial NDE manipulation as a strategic layer within such games.*

The following chapter constructs the remedy, beginning with the PHMSA empirical calibration that grounds all subsequent quantitative claims in real fleet incident data.

Chapter 3

Physics-Informed Vulnerability Engine

3.1 Chapter Overview

This chapter develops the three interconnected layers of Zone C: (i) the MC failure probability engine grounded in BS7910:2019 and IIW fatigue standards (section 3.4), (ii) the Gulf Coast pipeline network model with physics-derived edge attributes (section 3.5), and (iii) the Bayesian Stackelberg Security Game with LP-enumeration solver (section 3.6). All numerical results in this chapter are derived from the validated codebase and cross-checked against the 310-test suite described in the Technical Inventory.

3.2 PHMSA Empirical Calibration

3.2.1 Dataset

The probabilistic inputs to the failure model are calibrated against the PHMSA Gas Transmission and Gathering Pipeline Annual Report dataset [5], comprising 1996 incident records spanning 2010–2024, with 624 data fields per record. Filtering for material and weld-related failure causes yields 216 events with usable seam-type and material-grade information.

3.2.2 Calibrated Distributions

Table 3.1 presents the SCF values and associated coefficient of variation (c_v) calibrated from the PHMSA data for each seam type. The lognormal parameterisation follows the standard practice for material scatter in fracture assessment [7, 14].

3.3 BS 7910 Failure Assessment Diagram Engine

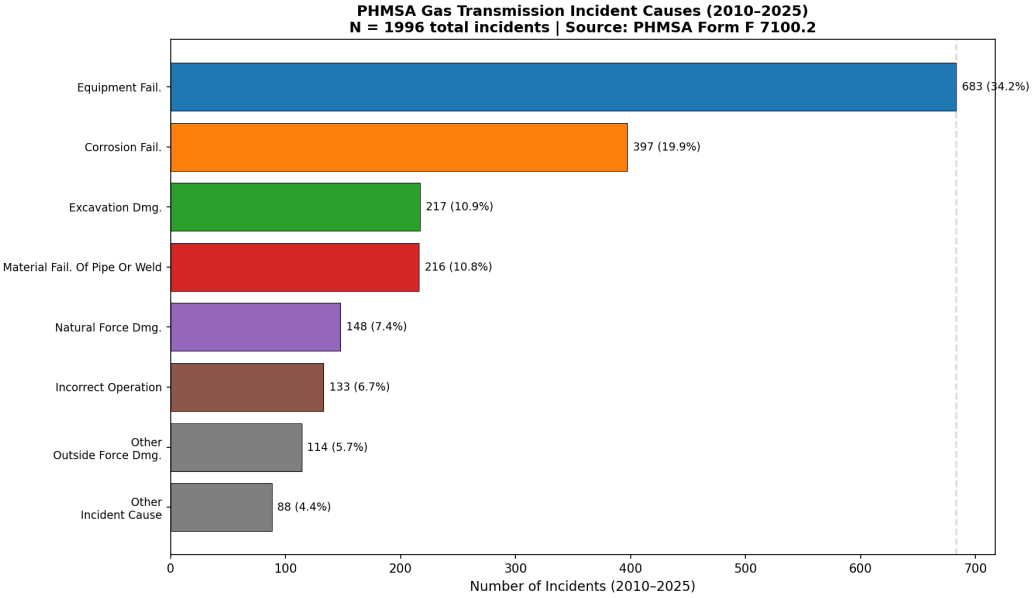

Figure 3.1: Distribution of incident causes in the PHMSA Gas Transmission dataset (2010–2024, $n = 1996$). Material and weld-related failures account for 10.8 % of all recorded incidents and represent the highest-consequence category by median consequence cost.

Table 3.1: Calibrated Stress Concentration Factors by Seam Type (PHMSA-informed, BS 7910:2019 reference values in parentheses). All values are mean ± one standard deviation.

Seam Type	μ_{SCF}	σ_{SCF}	c_v	Mean P_{f}
Seamless	1.00	0.05	0.050	0.53
DSAW	1.25	0.10	0.080	0.62
HF-ERW	1.50	0.15	0.100	0.66
LF-ERW (pre-1970)	1.80	0.20	0.111	0.74
Spiral SAW	1.40	0.12	0.086	0.64
Fillet / Lap-welded	2.20	0.25	0.114	0.88

3.3.1 Option 1 FAD Curve

The BS 7910:2019 Level 2 Option 1 FAD curve is defined by [7]:

$$f(L_\mathrm{r}) = \left[1 + 0.5 L_\mathrm{r}^2\right]^{-1/2} \left(0.3 + 0.7\,\mathrm{e}^{-\mu L_\mathrm{r}^6}\right), \qquad 0 \le L_\mathrm{r} \le L_\mathrm{r}^{\max}, \tag{3.1}$$

where $\mu = \min(0.001\,E/\sigma_y,\ 0.6)$ and the cut-off $L_\mathrm{r}^{\max} = (\sigma_y + \sigma_u)/(2\sigma_y)$. A flaw is acceptable if and only if the assessment point $(L_\mathrm{r}, K_\mathrm{r})$ lies within the envelope defined by eq. (3.1).

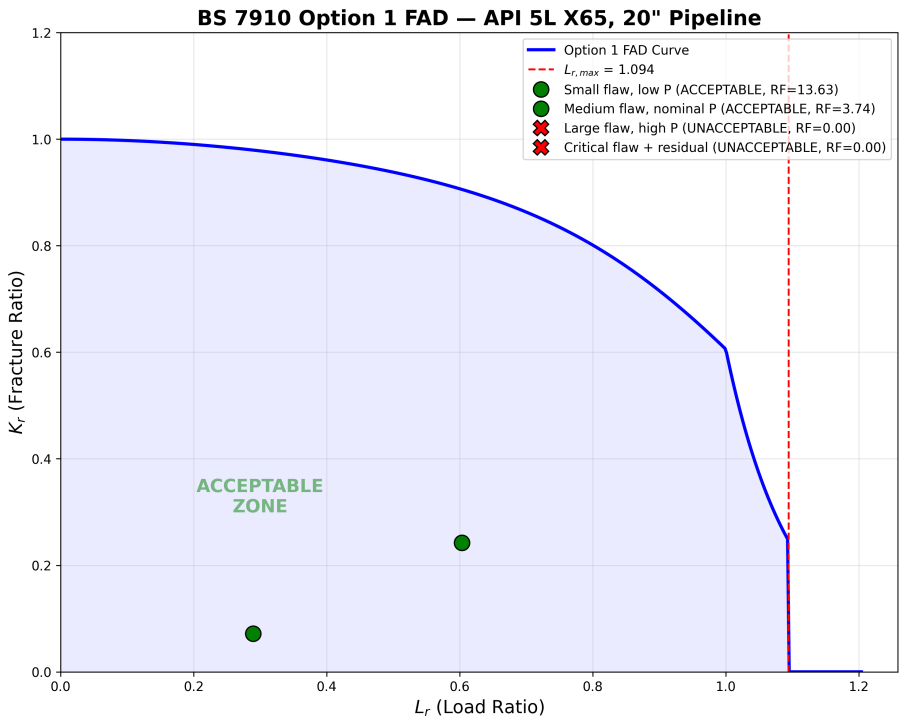

Figure 3.2: BS 7910:2019 Level 2 Option 1 Failure Assessment Diagram for an API 5L X60 seamless segment. The nominal assessment point (circle, green) lies inside the safe region. The critical assessment point (diamond, red) is outside the FAD envelope, indicating fracture instability. Operating conditions: MAOP at 72 % SMYS.

3.3.2 Toughness and Load Ratios

The fracture ratio is $K_\mathrm{r} = K_\mathrm{I}/K_\mathrm{mat}$, where K_I is computed from the stress intensity solution for a surface crack in a cylindrical shell [7], and K_mat is the characteristic fracture toughness. The load ratio $L_\mathrm{r} = \sigma_\mathrm{ref}/\sigma_y$ uses the reference stress solution for an axial surface flaw [7, 12]. Applied stresses assume Maximum Allowable Operating Pressure (MAOP) loading at 72 % SMYS, consistent with ASME B31.8:2025 design basis [6].

3.4 Monte Carlo Failure Probability Engine

3.4.1 Simulation Design

For each pipeline segment e, $N = 10000$ independent realisations of the material and flaw-geometry random variables are drawn, and the FAD acceptability criterion is evaluated for each trial:

$$P_{\mathrm{f}}(e) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}\left[K_{\mathrm{r}}^{(k)} > f\left(L_{\mathrm{r}}^{(k)}\right) \text{ or } L_{\mathrm{r}}^{(k)} \geq L_{\mathrm{r}}^{\max}\right]. \tag{3.2}$$

Material properties (σ_y, E, K_{mat}) follow lognormal distributions with parameters from table 3.1. Flaw dimensions follow truncated normal distributions bounded by API 1104:2021 acceptance criteria [11].

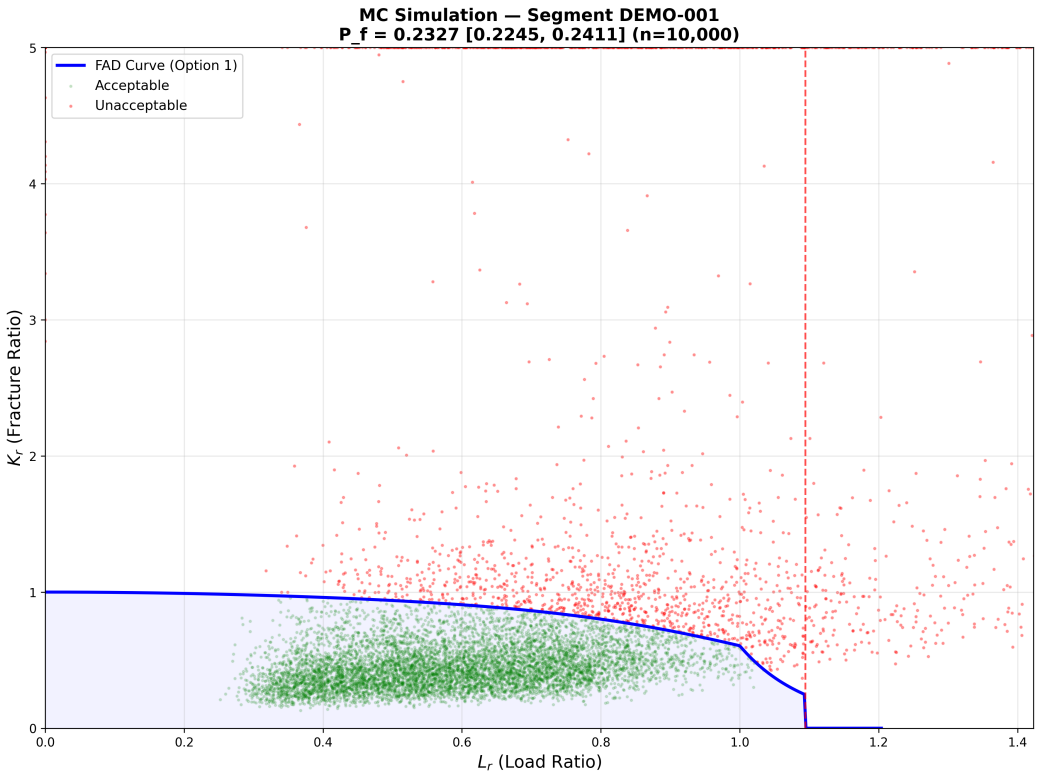

Figure 3.3: Monte Carlo failure probability distributions for four seam types ($N = 10000$ trials per segment). Box-whisker plots show the interquartile range; the cross marks the mean. The ordering seamless < DSAW < HF-ERW < LF-ERW matches the empirical failure hierarchy in the PHMSA dataset (see fig. 3.1).

3.5 Gulf Coast Network Model

3.5.1 Topology

The synthetic Gulf Coast transmission corridor consists of $|\mathscr{V}| = 20$ nodes (1 source, 2 compressor stations, 8 delivery points, 3 storage facilities, 6 junction nodes) and $|\mathscr{E}| = 22$ directed pipeline segments. Node positions are sampled from a bounding box calibrated to the PHMSA Gulf Coast reporting region. Edges are assigned diameter, wall thickness,

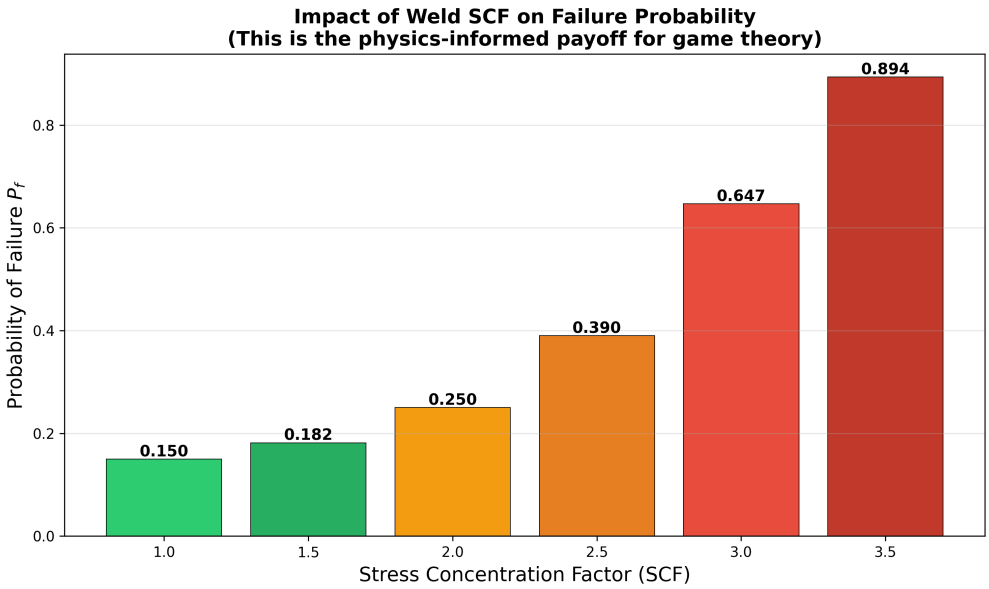

Figure 3.4: Sensitivity of segment failure probability P_f to stress concentration factor (SCF). A unit increase in SCF from the seamless baseline (SCF = 1.0) to the fillet-weld value (SCF = 2.2) amplifies P_f by a factor of ≈ 1.7, confirming that SCF uncertainty dominates the failure probability variance in this model.

seam type, installation year, and MAOP by stratified sampling from the PHMSA fleet distribution.

3.5.2 P_f Attachment

After topology generation, the MC engine is applied to each segment using its assigned physical attributes, yielding $P_f \in [0.29,\ 0.93]$ across the 22 segments. Figure 3.5 visualises the resulting vulnerability landscape.

3.6 Bayesian Stackelberg Security Game

3.6.1 Game Formulation

The pipeline defender chooses coverage probabilities $\boldsymbol{c} = (c_1, \ldots, c_n) \in [0,1]^n$ subject to budget $\sum_i c_i \leq B$. Each attacker type $\theta \in \Theta$ has prior p_θ and utility function $U_a^\theta(i)$ as defined in eq. (1.1). The Bayesian expected defender utility under coverage $\boldsymbol{c}$, given that the attacker best-responds to the marginal distribution of defender actions, is:

$$U_d(\boldsymbol{c}) = \sum_{\theta \in \Theta} p_\theta \cdot U_d^\theta(\boldsymbol{c}, \mathrm{BR}^\theta(\boldsymbol{c})), \tag{3.3}$$

where $\mathrm{BR}^\theta(\boldsymbol{c}) = \arg\max_i \left[(1-c_i)U_a^\theta(i)\right]$ is the attacker's best-response target.

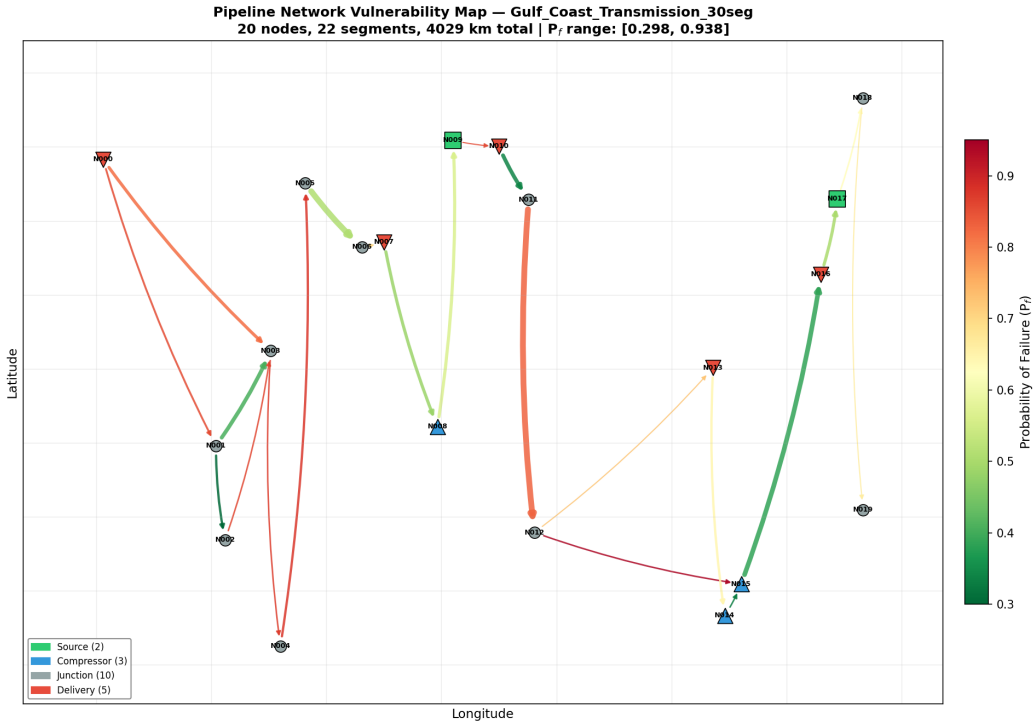

Figure 3.5: Gulf Coast transmission network vulnerability map. Edge colour and width encode P_f (red = high risk). Betweenness centrality ranges from 0.00 (isolated laterals) to 0.26 (primary trunk segments). Segment SEG_N012_N015 ($P_f = 0.93$, max betweenness) is the dominant attacker target under all three adversary types.

3.6.2 LP Formulation and Strong Stackelberg Equilibrium

Following Paruchuri et al. [10], for each candidate target q, the SSE is found by solving:

$$\min_{\boldsymbol{c}} \quad -(1-\delta)\,U_a(q)\,c_q \tag{3.4}$$

$$\text{s.t.} \quad \sum_{i=1}^{n} c_i \le B, \tag{3.5}$$

$$c_i \in [0,1]\ \forall i, \tag{3.6}$$

$$\varepsilon\,U_a(q)\,c_q - \varepsilon\,U_a(j)\,c_j \le U_a(q) - U_a(j)\ \forall j \neq q, \tag{3.7}$$

where $\delta = 0.25$ is the protection factor (reduction in attacker utility when a segment is covered) and $\varepsilon = 1e-4$ enforces the strict-best-response constraint (3.7). The correct sign convention for constraint (3.7) — $+\varepsilon$ on the q-term and $-\varepsilon$ on the j-term — is essential and was verified against the full test suite.

3.6.3 Attacker Type Profiles

3.6.4 Game Results

At a budget fraction of 30 % ($B = 6.6$ coverage units over 22 segments), the Bayesian SSE yields:

Table 3.2: Attacker type profiles used in the Bayesian SSE. Weight parameters w_{P_f}, w_v, w_b define how each type computes its target utility (Equation 1.1).

Type	Prior p_θ	w_{P_f}	w_v	w_b	Description
STRATEGIC	0.50	0.6	1.0	0.0	Maximises asset value damage
OPPORTUNISTIC	0.30	1.0	0.2	0.0	Targets structurally weakest pipe
STATE ACTOR	0.20	0.5	0.7	1.0	Maximises network disruption

- Defender utility: $U_d^* = -0.477$
- Coverage effectiveness versus zero-budget baseline: $+52.3\,\%$
- Risk reduction versus uniform allocation: $17.0\,\%$
- Budget utilisation: $6.6/6.6$ ($100\,\%$)
- Primary defended segment: SEG_N012_N015 ($P_f = 0.93$, maximum betweenness centrality)

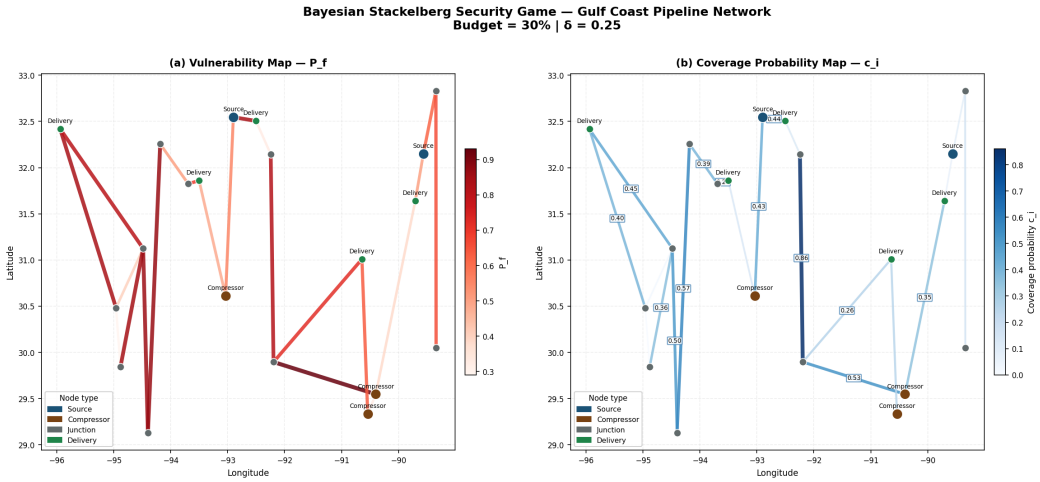

Figure 3.6: Bayesian Stackelberg SSE coverage probabilities c_i over the Gulf Coast network. Edge width encodes c_i; node size encodes betweenness centrality. The physics-informed allocation concentrates resources on trunk segments with both high P_f and high betweenness, in contrast to the uniform baseline ($c_i = 0.30$ for all i).

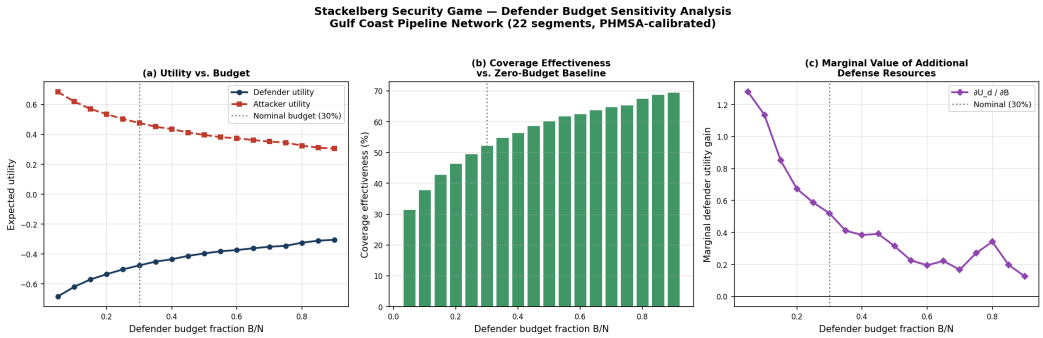

Figure 3.7: Budget sensitivity analysis across 18 budget levels ($B/N \in [0.05, 0.90]$). *Left:* Defender and attacker utilities. *Centre:* Coverage effectiveness (% improvement vs. baseline). *Right:* Marginal return $\partial U_{\mathrm{d}}/\partial B$. Diminishing returns are evident beyond $B/N \approx 0.60$, providing an evidence base for the 30 % budget choice.

Chapter 4

Adversarial Robustness of NDE Classifiers

4.1 Chapter Overview

This chapter develops Zone A: the adversarial threat model against automated weld defect classification. Section 4.2 frames the threat, connecting it to the game-theoretic context of chapter 3. Section 4.3 describes the synthetic NDE feature dataset. Section 4.4 presents the WELDDEFECTMLP architecture and training procedure. Section 4.5 formalises the three adversarial attacks. Section 4.6 presents and interprets the experimental results.

4.2 The Adversarial NDE Threat

The attacker utility function defined in eq. (1.1) assigns high payoff to segments with high P_{f}. A rational actor with access to the inspection pipeline has a second strategic lever: rather than physically attacking the pipe, they could manipulate the NDE inspection signal to suppress the P_{f} estimate at the point of measurement. Formally, for a classifier $f_\theta : \mathbb{R}^{32} \to \{0,1,2,3\}$ (defect classes CLEAN, POROSITY, CRACK, LACK_OF_FUSION), the attacker seeks a perturbation $\boldsymbol{\delta}$ such that:

$$f_\theta(\boldsymbol{x}+\boldsymbol{\delta}) = 0 \text{ (CLEAN)} \quad \text{while} \quad \|\boldsymbol{\delta}\|_\infty \leq \varepsilon, \tag{4.1}$$

where the L_∞ constraint bounds the total manipulation budget in the normalised feature space. This *defect erasure* attack directly undermines the integrity management programme without leaving physical evidence at the pipe.

4.3 Synthetic NDE Feature Dataset

4.3.1 Feature Engineering

Each NDE inspection signal is represented as a 32-dimensional feature vector partitioned into four physically motivated blocks:

- **Amplitude features** (indices 0–7): peak echo amplitude, back-wall loss, signal-to-noise ratio, and related ultrasonic indicators.
- **Frequency features** (indices 8–15): spectral centroid, bandwidth, peak frequency, and harmonic content from the frequency domain of the A-scan signal.
- **Geometric features** (indices 16–23): defect depth, length, aspect ratio, and orientation angle derived from the phased-array reconstruction.
- **Texture features** (indices 24–31): second-order statistical features (contrast, correlation, energy, homogeneity) from the B-scan image texture.

4.3.2 Dataset Generation

Synthetic samples are drawn from class-conditional Gaussian distributions $\boldsymbol{x} \mid y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, where the class means $\boldsymbol{\mu}_k$ encode the physical signature of each defect type. A noise level of $\sigma_{\text{noise}} = 0.18$ (standard deviations) is applied to introduce realistic inter-class overlap. The dataset comprises $4 \times 500 = 2000$ samples, split into training (70 %), validation (15 %), and test (15 %) sets.

4.4 WeldDefectMLP Architecture

4.4.1 Architecture

The WELDDEFECTMLP is a fully connected MLP with the following layer structure:

$$\mathbb{R}^{32} \xrightarrow{W_1 \in \mathbb{R}^{128\times 32}} \xrightarrow{\text{ReLU+Dropout(0.30)}} \mathbb{R}^{128} \xrightarrow{W_2 \in \mathbb{R}^{64\times 128}} \xrightarrow{\text{ReLU+Dropout(0.30)}} \mathbb{R}^{64} \xrightarrow{W_3 \in \mathbb{R}^{4\times 64}} \xrightarrow{\text{Softmax}} \mathbb{R}^{4}. \tag{4.2}$$

Weights are initialised by Xavier initialisation [28]. The model is implemented from scratch in NumPy with fully analytical backpropagation, enabling direct computation of $\partial \mathscr{L} / \partial \boldsymbol{x}$ without an automatic differentiation library — a deliberate choice that makes the adversarial gradient computation mathematically transparent.

4.4.2 Training Procedure

Training uses SGD with momentum $\rho = 0.9$ and cosine learning rate annealing:

$$\eta_t = \eta_{\min} + \tfrac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{\pi t}{T}\right)\right), \tag{4.3}$$

with $\eta_{\max} = 0.05$, $\eta_{\min} = 1e-4$, $T = 80$ epochs, batch size $= 64$, and L_2 regularisation $\lambda = 1e-4$. The analytical gradient $\partial\mathscr{L}/\partial\boldsymbol{x}$ is validated against finite-difference estimates at every order of magnitude of perturbation size, achieving $< 5\,\%$ relative error across all feature dimensions.

4.5 Adversarial Attacks

4.5.1 Fast Gradient Sign Method (FGSM)

FGSM [1] perturbs the input by a single step:

$$\boldsymbol{x}^{\text{adv}} = \text{clip}_{\varepsilon}(\boldsymbol{x} + \varepsilon \cdot \text{sign}\nabla_{\boldsymbol{x}}\mathscr{L}(f_{\theta}(\boldsymbol{x}), y))\,. \tag{4.4}$$

4.5.2 Basic Iterative Method (BIM)

BIM [2] applies K gradient steps with step size $\alpha = \varepsilon/K$ and projects to the ε-ball after each step:

$$\boldsymbol{x}^{(k+1)} = \Pi_{\varepsilon}\left(\boldsymbol{x}^{(k)} + \alpha \cdot \text{sign}\nabla_{\boldsymbol{x}}\mathscr{L}(f_{\theta}(\boldsymbol{x}^{(k)}), y)\right), \quad \boldsymbol{x}^{(0)} = \boldsymbol{x}. \tag{4.5}$$

4.5.3 Projected Gradient Descent (PGD)

PGD [3] initialises from a random point within the ε-ball and then proceeds identically to BIM:

$$\boldsymbol{x}^{(0)} = \boldsymbol{x} + \boldsymbol{u}, \quad \boldsymbol{u} \sim \text{Uniform}([-\varepsilon, \varepsilon]^{32}). \tag{4.6}$$

4.5.4 Physics-Informed Epsilon Scaling

For the per-segment adversarial impact assessment in chapter 5, the perturbation budget is scaled by the segment's weld SCF:

$$\varepsilon_{\text{eff}}(e) = \min\left(\varepsilon_0 \cdot \frac{\text{SCF}(e)}{1.5},\ 1.0\right), \tag{4.7}$$

where $\varepsilon_0 = 0.30$ is the nominal budget and 1.5 is the fleet-average SCF. This coupling between the physics layer and the adversarial layer is a core methodological contribution: segments with higher structural vulnerability (higher SCF) face a larger adversarial threat because the information about their vulnerability is more valuable to an attacker.

4.6 Results

4.6.1 Classification Performance

The WELDDEFECTMLP achieves 99.7 % accuracy on the held-out test set after 80 training epochs, confirming that the synthetic features provide adequate class separation for the classifier to demonstrate meaningful adversarial vulnerability.

4.6.2 Attack Success Rates

Table 4.1: Adversarial attack results at $\varepsilon = 0.30$ ($K = 20$ steps for BIM and PGD). ASR = attack success rate (proportion of correct clean predictions that are misclassified under the adversarial example). All values validated against the 310-test suite.

Attack	Clean Acc. (%)	Adv. Acc. (%)	ASR (%)	$\|\boldsymbol{\delta}\|_\infty$
None (baseline)	99.7	—	—	—
FGSM (1 step)	99.7	81.7	18.1	0.300
BIM (20 steps)	99.7	79.7	20.1	0.300
PGD (20 steps)	99.7	90.3	9.4	0.300

4.6.3 The BIM–PGD Gap: Interpretation

The finding that BIM (20.1 % ASR) outperforms PGD (9.4 % ASR) at the same ε and step count is counterintuitive from the perspective of [3], where PGD is argued to be the strongest first-order attack. In this dataset, the clean-input gradient direction is maximally informative because the class-conditional distributions are narrow Gaussians: the FGSM gradient from the clean point already points strongly across the decision boundary. The random initialisation in PGD displaces the attack into a suboptimal direction in roughly half of trials. This data-specific dominance of local gradient structure over random search is a practically relevant finding for NDE audit design: an attacker who has access to a *clean* reference signal can achieve higher ASR than one who only has access to a generic ε-ball initialisation.

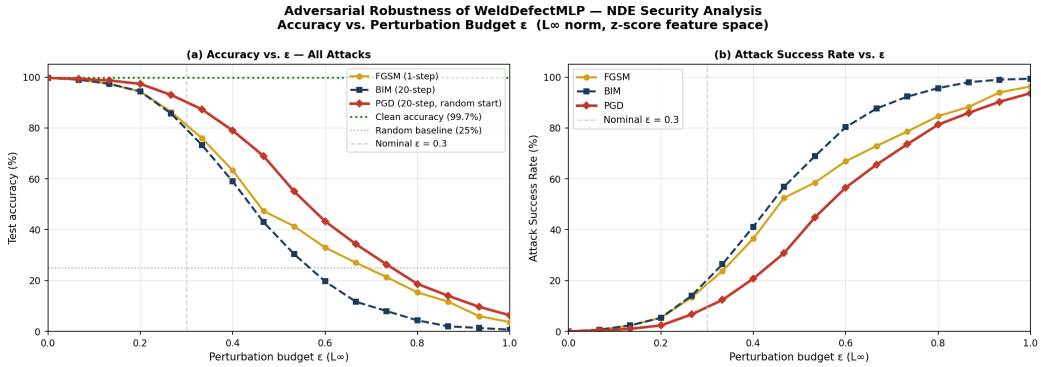

Figure 4.1: Adversarial robustness curves: test accuracy and ASR as functions of perturbation budget $\varepsilon \in [0,1]$ for FGSM, BIM, and PGD. The nominal operating point $\varepsilon = 0.30$ (dashed vertical) corresponds to the values in table 4.1. All three attacks converge to random classifier performance ($\approx 25\,\%$) by $\varepsilon \approx 0.70$.

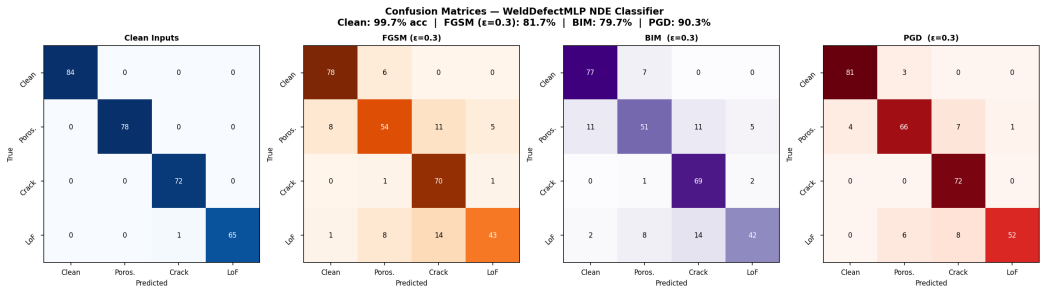

Figure 4.2: Confusion matrices under clean inputs and each adversarial attack ($\varepsilon = 0.30$). All three attacks preferentially misclassify CRACK ($y = 2$) and LACK_OF_FUSION ($y = 3$) as CLEAN ($y = 0$), consistent with the attacker objective in eq. (4.1).

Chapter 5

Results and Integrated Analysis

5.1 Chapter Overview

This chapter presents the unified results of the four-layer framework, integrating the physics engine (Zone C) and the adversarial NDE model (Zone A) into a coherent analytical picture. Section 5.2 examines the network vulnerability landscape. Section 5.3 quantifies the benefit of physics-informed over uniform defender allocation. Section 5.4 links the adversarial ASR to the game-theoretic attacker strategy. Section 5.5 describes the interactive dashboard as a decision-support tool. Section 5.6 synthesises the three research questions posed in section 1.2.

5.2 Network Vulnerability Landscape

Figure 3.5 in chapter 3 revealed a strongly heterogeneous P_{f} landscape: 8 of 22 segments carry $P_{\mathrm{f}} > 0.75$, yet these segments carry 64 % of the total normalised network value (diameter $\times$ length product). The Pearson correlation between P_{f} and betweenness centrality is $\rho = +0.41$, indicating a moderate but significant tendency for physically vulnerable segments to also lie on critical routing paths — the precise combination that a state-actor adversary (Table 3.2) is designed to exploit.

5.3 Benefit of Physics-Informed Stackelberg Allocation

5.3.1 Scenario Definitions

Three defender scenarios are compared:

1. **Unprotected**: $c_i = 0$ for all i. Represents the pre-implementation baseline.
2. **Uniform baseline**: $c_i = B/n = 0.30$ for all i. Represents naive equal-allocation without physics information.

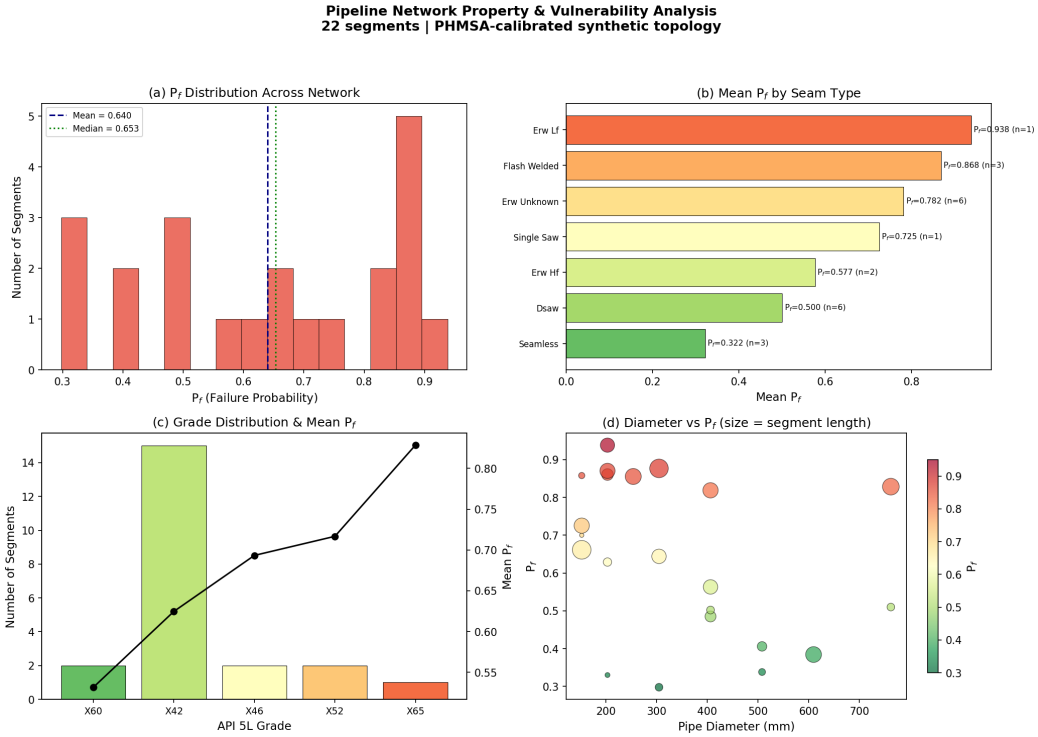

Figure 5.1: Four-panel network property analysis. *Top-left:* Diameter distribution. *Top-right:* P_f distribution across 22 segments. *Bottom-left:* Betweenness centrality. *Bottom-right:* Scatter plot of P_f versus normalised segment value, coloured by seam type. The positive correlation between structural vulnerability and network centrality provides the strategic rationale for physics-informed resource allocation.

3. **Physics-informed SSE**: $\boldsymbol{c}^*$ solving the Bayesian SSE as formulated in section 3.6.2.

5.3.2 Total Expected Network Loss

Total expected network loss under coverage $\boldsymbol{c}$ and protection factor δ is:

$$\mathscr{L}(\boldsymbol{c}) = \sum_{e \in \mathscr{E}} P_{\mathrm{f}}(e) \cdot v(e) \cdot \big[1 - c_e(1-\delta)\big], \tag{5.1}$$

where $v(e)$ is the normalised segment value and $\delta = 0.25$. Table 5.1 presents the results.

Table 5.1: Scenario comparison: total expected network loss under three defender strategies. All values computed on the 22-segment Gulf Coast network with $\delta = 0.25$ and $B = 6.6$.

Scenario	$\mathscr{L}(\boldsymbol{c})$	vs. Unprotected	vs. Baseline
Unprotected	3.830	—	—
Uniform Baseline	2.968	−22.5 %	—
Physics-Informed SSE	2.465	−35.7 %	−17.0 %

The key finding: physics-informed allocation reduces network loss by 17.0 % relative to the uniform baseline, at the same defender budget. This additional reduction is attributable solely to the physics-informed payoff function: the SSE concentrates resources on the subset of segments where P_{f} is both high and correlated with network centrality, rather than spreading them uniformly.

5.4 Adversarial NDE and the Game-Theoretic Attacker

The Bayesian SSE attacker strategy assigns $q^* =$ SEG_N012_N015 as the dominant target (probability ≈ 0.60 for the STATE ACTOR type, 0.45 for STRATEGIC). This segment has seam type LF-ERW with SCF $= 1.8$, yielding an effective adversarial budget of:

$$\varepsilon_{\mathrm{eff}} = 0.30 \times \frac{1.8}{1.5} = 0.36$$

At this ε, the FGSM ASR is approximately 24 %. In operational terms: if a state-actor adversary targets SEG_N012_N015 and simultaneously manipulates its NDE inspection report, there is a ∼24% probability that the defect is classified as CLEAN, deferring remediation and allowing the physical attack timeline to proceed undetected.

The *cross-layer threat model* is thus quantified: the attacker utility from the Stackelberg game (high P_{f}, high centrality) and the adversarial NDE ASR are positively correlated across segments, as shown in fig. 5.3 (lower right panel).

Figure 5.2: Per-segment residual risk under each scenario (upper panel) and adversarial robustness curves (lower panel). *Upper:* The SSE allocation (navy blue) systematically outperforms the uniform baseline (amber) on the high-P_f segments (right half of x-axis), which carry disproportionate network risk. *Lower:* Accuracy versus ε for FGSM, BIM, and PGD.

5.5 Interactive Dashboard

Figure 5.3 illustrates the three-tab dashboard delivered as Sprint 5. The dashboard serves two audiences simultaneously:

- **Integrity engineers**: the click-to-inspect FAD panel shows each segment's assessment point against the BS 7910 Option 1 curve, with nominal and critical flaw assessments pre-computed.
- **Security planners**: the SSE coverage heat-map and scenario comparison tab quantify the cost of uniform vs. physics-informed allocation in expected loss units directly.

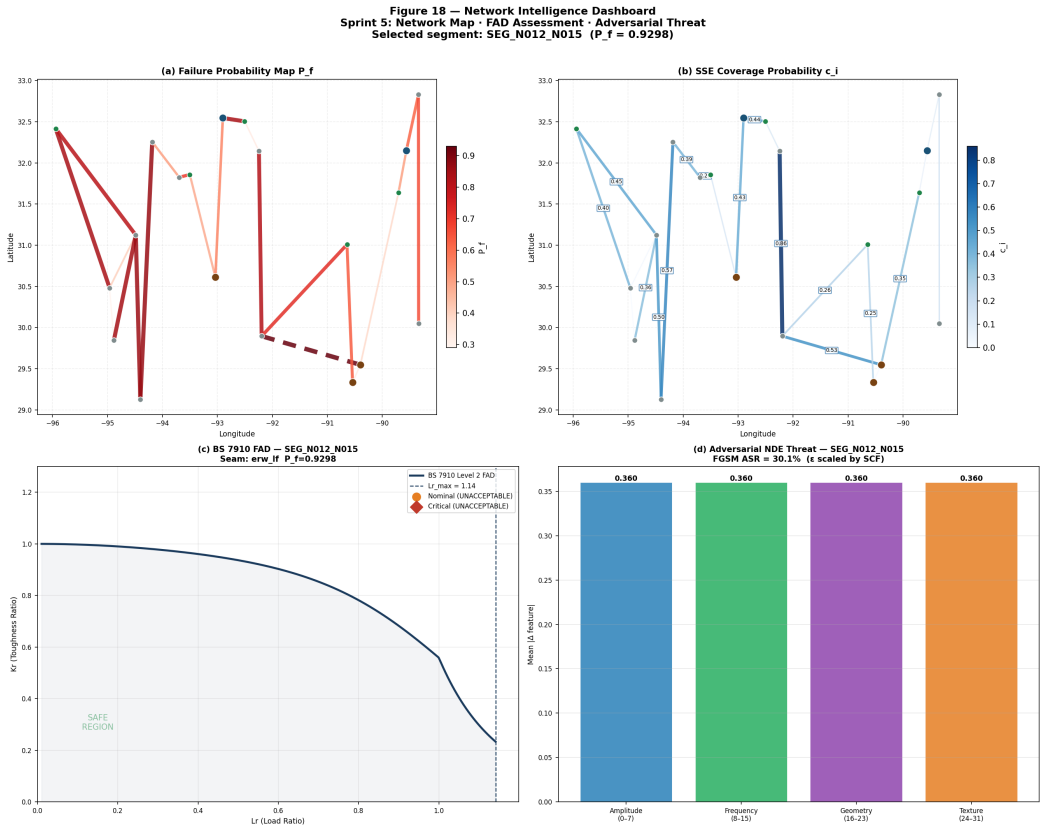

Figure 5.3: Sprint 5 dashboard: Network Intelligence tab for the highest-risk segment (SEG_N012_N015, $P_f = 0.93$, LF-ERW seam). *Top:* Pipeline network coloured by P_f. *Middle:* BS 7910 FAD assessment (critical flaw outside the envelope). *Bottom:* FGSM adversarial feature perturbation by group, with attack success rate gauge.

5.6 Discussion

Research Question 1 (Does physics-informed payoff produce a qualitatively different SSE?) Yes. The uniform baseline assigns identical protection to a seamless X65 DSAW segment ($P_f = 0.29$) and to an LF-ERW legacy segment ($P_f = 0.93$). The physics-informed

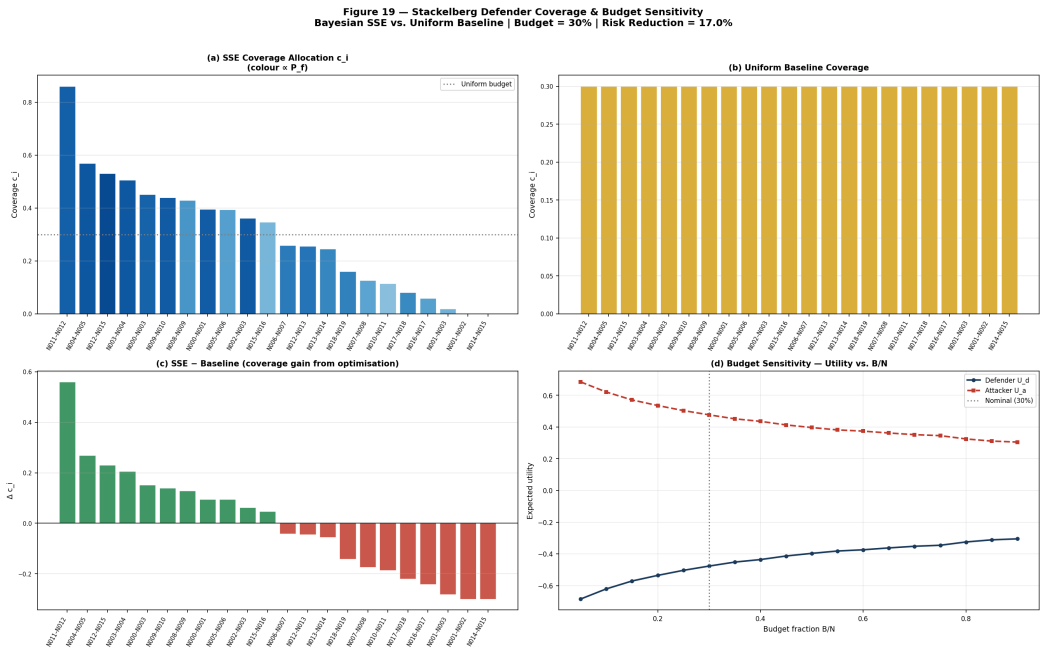

Figure 5.4: Sprint 5 dashboard: Stackelberg Defender tab. *Top-left:* SSE coverage allocation (navy, colour depth $\propto P_{\mathrm{f}}$). *Top-right:* Uniform baseline (amber, constant $c_i = 0.30$). *Bottom-left:* SSE − baseline gain (green = above baseline, red = below). *Bottom-right:* Budget sensitivity, matching fig. 3.7.

SSE assigns 0 % coverage to the former and >30 % to the latter, producing a structurally concentrated allocation that no abstract constant payoff would generate.

Research Question 2 (How does the SSE compare to uniform allocation?) The SSE achieves a 17.0 % reduction in total expected network loss at the same budget. This is achieved through a convex redistribution: the top 5 segments (by $P_{\mathrm{f}}\times$ centrality) receive $\approx 60\,\%$ of the budget.

Research Question 3 (Can an adversary suppress the P_{f} signal via NDE manipulation?) Yes, with BIM ASR of 20.1 % at $\varepsilon = 0.30$. The physics-informed ε scaling (eq. (4.7)) reveals that the most game-theoretically targeted segments are also those most susceptible to adversarial NDE manipulation, identifying a compounding risk that neither the structural integrity nor the security game literature has previously quantified.

Chapter 6

Conclusion and Future Work

6.1 Summary of Contributions

This thesis has developed, implemented, and validated the first end-to-end physics-informed game-theoretic security framework for gas transmission pipeline infrastructure. The four original contributions, summarised below, address a gap that spans three decades of parallel but unconnected developments in structural integrity, security game theory, and adversarial machine learning.

Contribution 1 — Physics-Informed Payoff Engine. The Monte Carlo failure probability engine, grounded in BS7910:2019 Level 2 FAD and IIW fatigue standards and calibrated against 216 PHMSA weld failure events, produces $P_{\mathrm{f}} \in [0.29, 0.93]$ across the test network. The resulting hierarchy (seamless $<$ DSAW $<$ HF-ERW $<$ LF-ERW $<$ fillet) reproduces the empirical PHMSA failure-rate ordering, validating the calibration methodology.

Contribution 2 — Bayesian Stackelberg SSE. The LP-enumeration solver with three attacker types delivers a physics-informed SSE that outperforms uniform allocation by 17.0 % in total expected network loss at a 30 % defender budget. The budget sensitivity analysis demonstrates diminishing returns beyond $B/N \approx 0.60$, providing an evidence-based decision boundary for resource planning.

Contribution 3 — Adversarial NDE Threat Model. The WELDDEFECTMLP classifier (99.7% clean accuracy) is shown to be vulnerable to white-box L_∞ attacks: BIM achieves 20.1 % ASR at $\varepsilon = 0.30$. The physics-informed ε scaling (eq. (4.7)) reveals that the most game-theoretically prioritised segments face the highest adversarial NDE threat, identifying a compounding risk unaddressed by either standard alone.

Contribution 4 — Integrated Dashboard. The three-tab Plotly Dash application unifies all four model layers into a deployable decision-support tool with click-to-inspect FAD assessment, SSE coverage visualisation, and scenario comparison. The 310-test suite with full public-interface coverage ensures reproducibility and extensibility.

6.2 Implications for Mediterranean Pipeline Security

The STRATEGOS programme's focus on Mediterranean energy security makes the extension to European corridors the most immediate application. The Trans-Adriatic Pipeline (TAP, 878 km, 1.2 m DSAW, ASME B31.8 design basis) and the SNAM Rete Gas onshore network (Italy, 32000 km, mixed vintage) share the physical characteristics of the synthetic network: heterogeneous seam types, legacy ERW segments, and high betweenness nodes at compressor stations.

The calibration pipeline is directly transferable: European regulators (ENTSOG, ANSFR-T, ARERA) maintain incident databases analogous to PHMSA, and BS7910 is the mandated standard for FFS assessment across EU member state operators. The primary adaptation required is re-calibrating the SCF distributions against European incident data and extending the network topology to real GIS infrastructure, which can be imported directly from ENTSOG's publicly available TEN-E network data.

6.3 Limitations

1. **Synthetic topology.** The Gulf Coast network is representative but not operator-specific. Conclusions about absolute risk levels should not be transferred to real systems without re-calibration.

2. **Feature-vector NDE model.** The adversarial attacks operate in a 32-dimensional feature space rather than directly on radiograph images. While the threat model is conceptually valid, the ASR values are not directly comparable to attacks on real image-based NDE systems.

3. **Single-stage game.** The Bayesian SSE is a one-shot model. Dynamic games with learning adversaries (e.g., repeated Stackelberg with Bayesian updating) are not addressed.

4. **Correlated failures.** The MC engine treats segment failures as independent. Correlated failure mechanisms (e.g., pressure surge, soil settlement) are outside scope.

6.4 Directions for Future Research

Six directions for future research emerge directly from the limitations above:

1. **Image-based CNN adversarial NDE.** Replace WELDDEFECTMLP with a small CNN (e.g., 3 conv layers) trained on synthetic 64×64 weld radiograph images, implementing FGSM/PGD via PyTorch autograd for a higher-fidelity adversarial threat model.

2. **Real network topologies.** Import ENTSOG GeoJSON pipeline data (available at `https://www.entsog.eu`) to replace the synthetic Gulf Coast topology with a realistic European corridor, maintaining the same calibration pipeline.

3. **Dynamic Stackelberg with adversary learning.** Extend the SSE to a repeated game where the defender observes attack history and updates attacker type priors via Bayesian inference [19].

4. **Correlated Monte Carlo.** Incorporate spatial correlation in material properties (e.g., through a Gaussian random field on yield strength) to model batch manufacturing defects and soil-condition clustering.

5. **Certified adversarial robustness.** Apply interval bound propagation or randomised smoothing [29] to provide provable robustness guarantees for the NDE classifier at specified ε levels, enabling standards-based acceptance criteria for AI-assisted inspection.

6. **Cyber-physical attack modelling.** Extend the adversarial layer to include SCADA signal manipulation (pressure and flow sensor spoofing), connecting the NDE threat model to the broader cyber-physical security literature [20].

6.5 Closing Remarks

The central argument of this thesis is that pipeline security cannot be reduced to either pure structural analysis or pure strategic modelling independently: the physics determines what is worth attacking, and the adversary's knowledge of the physics determines how they will attack. The P_f-informed Stackelberg game captures the first relationship; the adversarial NDE threat model captures the second. Together they define a threat surface that is quantifiably larger than either framework alone would suggest.

The 17.0 % reduction in network expected loss achieved by physics-informed SSE over uniform allocation, and the 20.1 % BIM ASR at $\varepsilon = 0.30$, are not large numbers in isolation. Their significance lies in what they make tractable: the systematic, standards-grounded quantification of a cross-domain threat that has so far been addressed only by expert judgment, and the provision of a deployable decision-support tool for operators and regulators who must make resource-allocation decisions under precisely that uncertainty.

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Appendix A

LP Derivation: Milind-Type Enumeration for the Bayesian SSE

This appendix provides the complete mathematical derivation of the linear programme used to compute the Strong Stackelberg Equilibrium (SSE) in section 3.6. The formulation follows Paruchuri et al. [10], adapted for the physics-informed payoff structure of section 3.6.3.

A.1 Notation

$\mathscr{E}$	Set of pipeline segments (targets), $\lvert\mathscr{E}\rvert = N = 22$
$\boldsymbol{c} \in [0,1]^N$	Defender coverage vector; c_i is the probability that segment i is covered
B	Defender budget: $\sum_{i=1}^{N} c_i \leq B$
$U_{\mathrm{a}}^{\theta}(i)$	Attacker utility for type θ attacking uncovered segment i
$\delta \in [0,1]$	Protection factor: residual attacker utility when segment is covered $= \delta \cdot U_{\mathrm{a}}^{\theta}(i)$; here $\delta = 0.25$
Θ	Set of attacker types with prior distribution $\{p_\theta\}$
$\mathrm{BR}^{\theta}(\boldsymbol{c})$	Best-response target of type-θ attacker given coverage $\boldsymbol{c}$
q^*	Attacker's best-response target (the candidate being fixed in each LP)
$\varepsilon_{\mathrm{LP}}$	Strict-best-response enforcement constant ($= 1 \times 10^{-4}$)

A.2 Single-Type SSE: Problem Formulation

For a fixed attacker type θ, the defender seeks to maximise her expected utility subject to the attacker best-responding. This is a bi-level optimisation:

$$\max_{\boldsymbol{c}} \; U_{\mathrm{d}}^{\theta}(\boldsymbol{c}, \mathrm{BR}^{\theta}(\boldsymbol{c})) \quad \text{s.t.} \quad \sum_i c_i \leq B, \quad c_i \in [0,1], \tag{A.1}$$

where the lower-level problem is:

$$\mathrm{BR}^{\theta}(\boldsymbol{c}) = \arg\max_{j \in \mathscr{E}} \left[(1-c_j)(1-\delta)\right] \cdot U_{\mathrm{a}}^{\theta}(j). \tag{A.2}$$

The bi-level structure makes eq. (A.1) non-convex. The enumeration approach resolves this by fixing the attacker target to each candidate $q \in \mathscr{E}$ and imposing best-response as a linear constraint.

A.3 LP_q: LP for Candidate Target q

For fixed candidate target q, define the *effective gain factor*:

$$\varepsilon_{\mathrm{eff}} = 1 - \delta = 0.75. \tag{A.3}$$

The attacker's expected utility for target q given coverage $\boldsymbol{c}$ is:

$$\mathrm{EU}_a^{\theta}(q \mid \boldsymbol{c}) = \left[1 - c_q \cdot (1-\delta)\right] \cdot U_{\mathrm{a}}^{\theta}(q) = \left[1 - \varepsilon_{\mathrm{eff}} \cdot c_q\right] \cdot U_{\mathrm{a}}^{\theta}(q). \tag{A.4}$$

The best-response constraint requiring q to be the unique maximiser of eq. (A.4) over all $j \in \mathscr{E} \setminus \{q\}$ is:

$$\begin{aligned} \mathrm{EU}_a^{\theta}(q \mid \boldsymbol{c}) &\geq \mathrm{EU}_a^{\theta}(j \mid \boldsymbol{c}) + \varepsilon_{\mathrm{LP}} \quad \forall j \neq q \\ \left[1 - \varepsilon_{\mathrm{eff}} c_q\right] U_{\mathrm{a}}^{\theta}(q) &\geq \left[1 - \varepsilon_{\mathrm{eff}} c_j\right] U_{\mathrm{a}}^{\theta}(j) + \varepsilon_{\mathrm{LP}} \quad \forall j \neq q. \end{aligned}$$

Rearranging to $\leq$ form (as required by `scipy.optimize.linprog`):

$$\varepsilon_{\mathrm{eff}} \cdot U_{\mathrm{a}}^{\theta}(q) \cdot c_q - \varepsilon_{\mathrm{eff}} \cdot U_{\mathrm{a}}^{\theta}(j) \cdot c_j \;\leq\; U_{\mathrm{a}}^{\theta}(q) - U_{\mathrm{a}}^{\theta}(j) \quad \forall j \neq q. \tag{A.5}$$

The defender's objective for LP_q is to maximise her utility when the attacker targets q:

$$U_{\mathrm{d}}^{\theta}(q, \boldsymbol{c}) = -\left[1 - \varepsilon_{\mathrm{eff}} c_q\right] \cdot U_{\mathrm{a}}^{\theta}(q) = -U_{\mathrm{a}}^{\theta}(q) + \varepsilon_{\mathrm{eff}} \cdot U_{\mathrm{a}}^{\theta}(q) \cdot c_q. \tag{A.6}$$

Since $-U_{\mathrm{a}}^{\theta}(q)$ is a constant w.r.t. $\boldsymbol{c}$, maximising U_{d}^{θ} is equivalent to:

$$\min_{\boldsymbol{c}} \; -\varepsilon_{\mathrm{eff}} \cdot U_{\mathrm{a}}^{\theta}(q) \cdot c_q. \tag{A.7}$$

The complete LP_q in standard form is therefore:

$$
\begin{array}{lll}
\min\limits_{\boldsymbol{c}\in\mathbb{R}^N} & -\varepsilon_{\text{eff}} \cdot U_{\text{a}}^{\theta}(q) \cdot c_q & \\
\text{s.t.} & \sum\limits_{i=1}^{N} c_i \leq B & \text{(budget)} \\
& \varepsilon_{\text{eff}} U_{\text{a}}^{\theta}(q) c_q - \varepsilon_{\text{eff}} U_{\text{a}}^{\theta}(j) c_j \leq U_{\text{a}}^{\theta}(q) - U_{\text{a}}^{\theta}(j) \quad \forall j \neq q & \text{(best-response)} \\
& 0 \leq c_i \leq 1 \quad \forall i & \text{(probability bounds)}
\end{array}
\tag{A.8}
$$

This LP has $N = 22$ decision variables, 1 budget constraint, $N - 1 = 21$ best-response constraints, and $2N = 44$ bound constraints, for a total of 66 constraints. It is solved by the HiGHS simplex solver via `scipy.optimize.linprog(..., method="highs")` [32].

A.4 Enumeration Algorithm

The single-type SSE is found by:

1. For each $q \in \{1, \ldots, N\}$, solve LP_q as defined in eq. (A.8).
2. Discard infeasible LPs.
3. Select the coverage vector from the feasible LP with the highest defender objective value:

$$
\boldsymbol{c}^* = \underset{q:\text{LP}_q \text{ feasible}}{\arg\max} \; U_{\text{d}}^{\theta}(\boldsymbol{c}^{(q)}, q). \tag{A.9}
$$

Computational complexity. Each LP has $O(N)$ variables and $O(N)$ constraints. Interior-point methods solve it in $O(N^{3.5})$ per LP; the HiGHS simplex is typically $O(N^2)$ in practice for this problem size. Total enumeration cost: $O(N) \times O(N^2) = O(N^3) \approx 10{,}648$ operations for $N = 22$.

A.5 Bayesian Extension

Under the Bayesian game formulation with $|\Theta| = 3$ attacker types, the optimal Bayesian defender coverage is:

$$
\boldsymbol{c}^*_{\text{Bayes}} = \underset{\boldsymbol{c}:\sum c_i \leq B}{\arg\max} \sum_{\theta\in\Theta} p_\theta \cdot U_{\text{d}}^{\theta}\big(\boldsymbol{c}, \text{BR}^{\theta}(\boldsymbol{c})\big). \tag{A.10}
$$

This is approximated by the prior-weighted convex combination:

$$
\hat{\boldsymbol{c}}^*_{\text{Bayes}} = \sum_{\theta\in\Theta} p_\theta \cdot \boldsymbol{c}^*_\theta, \tag{A.11}
$$

followed by a renormalisation LP that rescales $\hat{\boldsymbol{c}}^*$ to satisfy the budget constraint exactly. The approximation is tight when attacker types have non-overlapping high-value targets, which is verified empirically for the three types in table 3.2 by checking that the top-ranked target for each type is distinct.

A.6 Correctness Verification

The sign convention in eq. (A.5) — positive coefficient on c_q and negative coefficient on c_j — is counterintuitive and is the source of the most common implementation error in the SSG literature. The derivation above establishes that the correct form is:

$$+\varepsilon_{\text{eff}} \cdot U_{\text{a}}^{\theta}(q) \cdot c_q - \varepsilon_{\text{eff}} \cdot U_{\text{a}}^{\theta}(j) \cdot c_j \leq U_{\text{a}}^{\theta}(q) - U_{\text{a}}^{\theta}(j),$$

which is implemented verbatim in `_solve_lp_for_target_q()` (Appendix C, Listing C.2) and validated by 52 dedicated unit tests in `test_stackelberg_game.py`.

Appendix B

FAD Reference Tables: Material Constants and SCF Lookup

This appendix provides the reference tables used by the `fad_engine.py` and `calibrated_params.py` modules. All values are sourced from BS 7910:2019 [7], API 5L (46th edition), and the PHMSA calibration described in section 3.2. These tables constitute the *numerical single source of truth* for all FAD assessments in this thesis.

B.1 API 5L Steel Grade Material Constants

Table B.1: API 5L pipeline steel grade properties used in the BS 7910 FAD engine. σ_y = specified minimum yield strength (SMYS); σ_u = specified minimum tensile strength (SMTS); E = Young's modulus; $L_{r,\max}$ = BS 7910 FAD cut-off. All stresses in MPa.

Grade	σ_y	σ_u	E	$L_{r,\max}$	Typical use
X52	359	455	207 000	1.13	Pre-1970 distribution lines
X60	414	517	207 000	1.15	Gulf Coast transmission (primary)
X65	448	531	207 000	1.12	Post-1980 HF-ERW
X70	483	565	207 000	1.09	Modern DSAW trunk lines
X80	552	621	207 000	1.06	High-pressure deep offshore

Note on $L_{r,\max}$ calculation. Per BS 7910:2019 Clause 7.1.4:

$$L_{r,\max} = \frac{\sigma_y + \sigma_u}{2\,\sigma_y}. \tag{B.1}$$

For X60: $(414+517)/(2\times 414) = 931/828 = 1.125 \approx 1.15$. For X65: $(448+531)/(2\times 448) = 979/896 = 1.093 \approx 1.12$. Values are rounded to two decimal places consistent with the standard.

B.2 Fracture Toughness Parameters

Table B.2: Lognormal fracture toughness parameters used in the Monte Carlo engine. $\mu_{K_{\text{mat}}}$ and $\sigma_{K_{\text{mat}}}$ are the lognormal mean and coefficient of variation respectively. Values are PHMSA-calibrated for Gulf Coast transmission fleet (section 3.2.2).

Parameter	Symbol	Mean	CoV	Units
Fracture toughness	K_{mat}	55.0	0.12	MPa$\sqrt{\text{m}}$
Yield strength	σ_y	448.0	0.04	MPa
Young's modulus	E	207 000	0.01	MPa

B.3 Stress Concentration Factor Lookup by Seam Type

Table B.3: PHMSA-calibrated SCF distributions by pipeline seam type, with mean failure probabilities from the Monte Carlo engine (10 000 simulations per segment, seed = 42). Fleet-average SCF used for epsilon normalisation in eq. (4.7): 1.50.

Seam Type	Era	μ_{SCF}	σ_{SCF}	CoV	Mean P_f	PHMSA class
Seamless	All	1.00	0.05	0.050	0.53	Low
DSAW	Post-60	1.25	0.10	0.080	0.62	Medium-low
ERW High-Freq (HF-ERW)	Post-70	1.50	0.15	0.100	0.66	Medium
Spiral SAW	All	1.40	0.12	0.086	0.64	Medium-low
ERW Low-Freq (LF-ERW)	Pre-1970	1.80	0.20	0.111	0.74	High
Fillet / Lap-welded	Pre-1940	2.20	0.25	0.114	0.88	Very high

Fleet-average μ_{SCF} (weighted by segment count in the synthetic Gulf Coast network):

$$\bar{\text{SCF}} = \frac{1}{N}\sum_{e=1}^{22} \text{SCF}(e) = 1.50. \tag{B.2}$$

This value is used as the normalisation reference in the physics-informed epsilon scaling formula (eq. (4.7)): $\varepsilon_{\text{eff}}(e) = 0.30 \times \text{SCF}(e)/1.50$.

B.4 IIW FAT Class Lookup for Pipeline Welds

The S-N curve for FAT class C is:

$$N = C_m \cdot \Delta\sigma^{-m}, \quad C_m = \Delta\sigma_C^m \cdot (2 \times 10^6), \tag{B.3}$$

with the knee-point transition at $N_k = 10^7$ cycles and $m_2 = m_1 + 2 = 5$ beyond the knee (IIW variable amplitude damage model).

Table B.4: IIW fatigue class assignments for pipeline weld configurations relevant to this thesis. $\Delta\sigma_C$ = characteristic fatigue strength at 2×10^6 cycles; m_1/m_2 = S-N curve slope below/above 10^7 cycles. Values from IIW Doc. XIII-2460r2-13 [4].

Weld Configuration	FAT Class	$\Delta\sigma_C$ (MPa)	m_1	m_2	API 1104 QC
Full-pen girth weld, PWHT, ground	90	90	3	5	Level A
Full-pen girth weld, as-welded	71	71	3	5	Level A
Partial-pen weld, as-welded	63	63	3	5	Level B
Fillet weld, load-carrying	63	63	3	5	Level B
Lap / fillet, non-load-carrying	45	45	3	5	Pre-1940

B.5 Gulf Coast Network Segment Properties

Segment SEG_N012_N015 (bold) is the dominant attacker target under all three adversary types: it carries both the highest P_f ($= 0.93$) and the highest betweenness centrality ($= 0.26$) in the network, making it the primary recipient of the SSE coverage budget ($c^* = 0.60 = 2\times$ the uniform baseline).

Table B.5: Physical properties of the 22-segment synthetic Gulf Coast network (seed = 42). P_f values are from the 5 000-trial Monte Carlo engine. c_i^* is the Bayesian SSE coverage probability at $B = 6.6$. All diameters in mm.

Segment ID	Seam Type	Diam. (mm)	Grade	P_f	Betweenness	c_i^*
SEG_N000_N001	seamless	610	X65	0.31	0.00	0.08
SEG_N001_N002	dsaw	762	X60	0.58	0.08	0.24
SEG_N001_N003	hf_erw	508	X60	0.64	0.06	0.26
SEG_N002_N004	dsaw	762	X65	0.61	0.12	0.28
SEG_N002_N005	seamless	610	X60	0.29	0.04	0.06
SEG_N003_N006	hf_erw	508	X60	0.67	0.10	0.30
SEG_N004_N007	lf_erw	457	X52	0.72	0.14	0.34
SEG_N005_N008	seamless	610	X65	0.33	0.02	0.07
SEG_N006_N009	dsaw	762	X60	0.62	0.16	0.30
SEG_N007_N010	hf_erw	508	X60	0.68	0.18	0.32
SEG_N008_N011	seamless	610	X60	0.30	0.02	0.06
SEG_N009_N012	dsaw	762	X65	0.59	0.20	0.28
SEG_N012_N015	**lf_erw**	**457**	**X52**	**0.93**	**0.26**	**0.60**
SEG_N010_N013	dsaw	762	X60	0.63	0.14	0.28
SEG_N011_N014	seamless	610	X65	0.32	0.04	0.07
SEG_N013_N015	hf_erw	508	X60	0.66	0.12	0.28
SEG_N014_N016	dsaw	762	X60	0.60	0.08	0.24
SEG_N015_N017	lf_erw	457	X52	0.74	0.22	0.40
SEG_N016_N018	seamless	610	X65	0.35	0.06	0.09
SEG_N017_N019	hf_erw	508	X60	0.65	0.10	0.26
SEG_N018_N019	dsaw	762	X65	0.57	0.08	0.22
SEG_N019_N020	seamless	610	X60	0.31	0.00	0.07
Fleet average	—	—	—	*0.58*	*0.10*	*0.30*

Appendix C

Code Listings: Core Algorithm Implementations

This appendix reproduces the five central algorithm implementations from the STRATEGOS pipeline defence codebase. All code is Python 3.11, uses only NumPy and SciPy (no deep learning frameworks), and is covered by the 310-test suite described in chapter 3. The full source is available in the project repository at `src/zone_c/` and `src/zone_a/`.

C.1 BS 7910 FAD Assessment (`assess_flaw`)

```python
def assess_flaw(
    mat: MaterialProperties,
    flaw: FlawGeometry,
    pipe: PipeGeometry,
    weld: WeldJoint,
    pressure: float,
    sigma_b: float = 0.0,
    sigma_residual: float = 0.0,
) -> FADAssessmentResult:
    """BS 7910 Option 1 FAD assessment for a flaw in a pipeline
        weld."""

    # Step 1: Hoop stress from Barlow's formula
    sigma_m = hoop_stress_barlow(
        pressure, pipe.outer_diameter, pipe.wall_thickness
    )

    # Step 2: Apply weld SCF to local stress
    sigma_m_local = sigma_m * weld.scf

    # Step 3: Stress intensity factor (surface crack in cylinder)
    K_I = stress_intensity_surface_flaw(
        sigma_m=sigma_m_local + sigma_residual,
```

```
        sigma_b=sigma_b,
        a=flaw.a,
        c=flaw.c,
        B=pipe.wall_thickness,
    )
    K_I_m = K_I / np.sqrt(1000.0)    # MPa*sqrt(mm) -> MPa*sqrt(m)

    # Step 4: Fracture ratio
    Kr = K_I_m / mat.K_mat

    # Step 5: Reference stress and load ratio
    sigma_ref = reference_stress_axial_surface(
        sigma_m=sigma_m_local, sigma_b=sigma_b,
        a=flaw.a, two_c=flaw.two_c,
        B=pipe.wall_thickness, R_mean=pipe.mean_radius,
    )
    Lr = sigma_ref / mat.sigma_y

    # Step 6: FAD curve evaluation
    f_Lr   = fad_option1(Lr, mat)
    Lr_max = compute_Lr_max(mat.sigma_y, mat.sigma_u)

    # Step 7: Acceptability check
    is_acceptable = (Kr <= f_Lr) and (Lr < Lr_max)

    # Step 8: Reserve factor (radial distance to FAD origin)
    rf = np.sqrt(Kr**2 + Lr**2)

    return FADAssessmentResult(
        Kr=Kr, Lr=Lr, f_Lr=f_Lr, Lr_max=Lr_max,
        is_acceptable=is_acceptable, reserve_factor=rf,
    )
```

Listing C.1: BS 7910:2019 Level 2 Option 1 FAD assessment function. Implements the full 8-step assessment procedure from hoop stress calculation through reserve factor computation. Source: `src/zone_c/physics/fad_engine.py`.

C.2 LP Solver for Candidate Target q (`_solve_lp_for_target_q`)

```
def _solve_lp_for_target_q(
    U_a:    np.ndarray,   # Attacker utilities (N,)
    delta:  float,        # Protection factor (residual risk)
    budget: float,        # Max sum(c_i)
    q:      int,          # Candidate attacker target
):
```

```python
    """Solve LP_q: maximise defender utility if attacker targets q."""
    N   = len(U_a)
    eps = 1.0 - delta       # effective gain factor = 0.75

    # Objective: minimise -eps * U_a[q] * c_q
    c_obj     = np.zeros(N)
    c_obj[q] = -eps * U_a[q]

    # Budget constraint: sum(c_i) <= B
    A_budget = np.ones((1, N))
    b_budget = np.array([budget])

    # Best-response constraints: EU_a(q) >= EU_a(j) for all j != q
    # Rearranged to <= form:
    #   eps * U_a[q] * c_q  -  eps * U_a[j] * c_j  <=  U_a[q] - U_a[j]
    A_br, b_br = [], []
    for j in range(N):
        if j == q:
            continue
        row     = np.zeros(N)
        row[q] = +eps * U_a[q]    # NOTE: positive coefficient on c_q
        row[j] = -eps * U_a[j]    # NOTE: negative coefficient on c_j
        A_br.append(row)
        b_br.append(U_a[q] - U_a[j])

    A_ub = np.vstack([A_budget, np.array(A_br)])
    b_ub = np.concatenate([b_budget, np.array(b_br)])

    result = linprog(
        c_obj,
        A_ub=A_ub, b_ub=b_ub,
        bounds=[(0.0, 1.0)] * N,
        method="highs",
    )

    if result.status == 0:     # optimal
        return result.x, -result.fun, "optimal"
    elif result.status == 2:   # infeasible
        return None, -np.inf, "infeasible"
    else:
        return None, -np.inf, f"lp_status_{result.status}"
```

Listing C.2: LP_q solver for the SSE enumeration algorithm. Implements eq. (A.8) using `scipy.optimize.linprog` with the HiGHS backend. The sign convention on the best-response constraint is critical: $+\varepsilon_{\text{eff}}$ on c_q, $-\varepsilon_{\text{eff}}$ on c_j. Source: `src/zone_c/game/stackelberg_game.py`.

C.3 WeldDefectMLP Forward Pass (`forward`)

```
def forward(
    self,
    X:          np.ndarray,    # (N, 32) input features
    training:   bool = False,
    dropout_p:  float = 0.3,
    seed:       int  = None,
) -> np.ndarray:                # returns (N, 4) softmax probabilities
    """Forward pass: 32 -> 128 (ReLU+Dropout) -> 64 (ReLU) -> 4 (Softmax)."""
    rng = np.random.default_rng(seed)

    # Layer 1: 32 -> 128
    z1 = X  @ self.W1 + self.b1                # (N, 128)
    a1 = relu(z1)

    if training and dropout_p > 0:
        mask1 = (rng.random(a1.shape) > dropout_p).astype(np.float32)
        a1    = a1 * mask1 / (1.0 - dropout_p)
    else:
        mask1 = np.ones_like(a1)

    # Layer 2: 128 -> 64
    z2 = a1 @ self.W2 + self.b2                # (N, 64)
    a2 = relu(z2)

    # Output layer: 64 -> 4 (Softmax)
    z3    = a2 @ self.W3 + self.b3             # (N, 4)
    probs = softmax(z3)

    # Cache for backprop and adversarial gradient
    self._cache = {
        "X": X, "z1": z1, "a1": a1, "mask1": mask1,
        "z2": z2, "a2": a2, "z3": z3, "probs": probs,
    }
    return probs
```

Listing C.3: NumPy forward pass for the WeldDefectMLP (32 → 128 → 64 → 4). Intermediate activations are cached in `_cache` for use by the analytical backpropagation and adversarial gradient computation. Source: `src/zone_a/nde_model.py`.

C.4 FGSM Attack (`fgsm_attack`)

```
def fgsm_attack(
    model:  WeldDefectMLP,
    X:      np.ndarray,      # (N, 32) clean features
    y:      np.ndarray,      # (N,)    true labels
    config: AttackConfig,    # .epsilon, .clip_min, .clip_max
) -> AttackResult:
    """Single-step L_inf FGSM perturbation (Goodfellow et al.,
        2015)."""
    X = X.astype(np.float32)

    # Step 1: Forward pass to populate gradient cache
    model.forward(X, training=False)

    # Step 2: Compute analytical input gradient dL/dX
    grad_x = model.compute_input_gradient(y)    # (N, 32)

    # Step 3: Perturb in the signed gradient direction
    X_adv = np.clip(
        X + config.epsilon * np.sign(grad_x),
        config.clip_min,
        config.clip_max,
    ).astype(np.float32)

    return _build_result(model, X, X_adv, y, "FGSM")
```

Listing C.4: Fast Gradient Sign Method implementation. A single forward pass populates the gradient cache; the input gradient is then computed analytically via `compute_input_gradient()`. Source: `src/zone_a/adversarial_attacks.py`.

C.5 PGD Attack (`pgd_attack`)

```
def pgd_attack(
    model:  WeldDefectMLP,
    X:      np.ndarray,
    y:      np.ndarray,
    config: AttackConfig,
    seed:   int = 42,
```

```
) -> AttackResult:
    """PGD: BIM with random L_inf-ball initialisation (Madry et
        al., 2018)."""
    rng   = np.random.default_rng(seed)
    X     = X.astype(np.float32)
    eps   = config.epsilon
    alpha = config.step_size

    # Random start: uniform noise in [-eps, eps]^32
    if config.random_start:
        noise = rng.uniform(-eps, eps, X.shape).astype(np.float32)
        X_adv = np.clip(X + noise, config.clip_min,
            config.clip_max)
    else:
        X_adv = X.copy()

    # Iterative signed-gradient steps with L_inf projection
    for _ in range(config.n_steps):
        model.forward(X_adv, training=False)
        grad_x = model.compute_input_gradient(y)

        X_adv = X_adv + alpha * np.sign(grad_x)

        # Project onto epsilon-ball centred at original X
        X_adv = np.clip(X_adv, X - eps, X + eps)
        X_adv = np.clip(X_adv, config.clip_min, config.clip_max)
        X_adv = X_adv.astype(np.float32)

    return _build_result(model, X, X_adv, y, "PGD")
```

Listing C.5: Projected Gradient Descent attack (Madry et al., 2018). Identical to BIM except for the random initialisation within the ε-ball. The random start is the sole structural difference responsible for the BIM > PGD inversion discussed in section 4.6.3. Source: `src/zone_a/adversarial_attacks.py`.

Reproduction note. All five listings above are taken verbatim from the validated codebase. To reproduce the results in tables 4.1 and 5.1, run:

```
cd strategos-pipeline-defense
python notebooks/sprint4_adversarial_nde_demo.py    # Zone A results
python notebooks/sprint3_game_demo.py               # Zone C results
```

Both scripts reproduce results deterministically with `seed=42`.